CS XII

Computer Science XII — 30-Day Ultimate Cheat-Sheet Booklet

Complete Edition · Days 1–30 · Urdish (Roman Urdu + English) · New Sindh Curriculum 2025–27

Is booklet mein

1	Computer Systems (HCI)	Days 1–4 · p. 3
2	Computational Thinking & Algorithms	Days 5–9 · p. 7
3	Programming Fundamentals	Days 10–13 · p. 12
4	Data and Analysis	Days 14–15 · p. 16
5	Application & Impacts of Computing	Days 16–19 · p. 18
6	Entrepreneurship in the Digital Age	Days 20–23 · p. 22
★	Golden Blitz & Recaps	Days 24–27 · p. 26
M	Mock, Mark & Repair	Days 28–30 · p. 30

30-din ka naqsha (har din = 3 ghante)

Day 01 Teach	Day 02 Teach	Day 03 Teach	Day 04 Checkpoint	Day 05 Teach	Day 06 Teach
Day 07 Teach	Day 08 Teach	Day 09 Checkpoint	Day 10 Teach	Day 11 Teach	Day 12 Teach
Day 13 Checkpoint	Day 14 Teach	Day 15 Checkpoint	Day 16 Teach	Day 17 Teach	Day 18 Teach
Day 19 Checkpoint	Day 20 Teach	Day 21 Teach	Day 22 Teach	Day 23 Checkpoint	Day 24 Golden blitz
Day 25 Golden blitz	Day 26 Golden blitz	Day 27 Recap	Day 28 Mock	Day 29 Mark	Day 30 Repair

Boxes ka matlab

Def (paper mein yehi likho): *Woh definition jo paper mein bilkul aise hi likhni hai.*

🧠 **Yaad rakho:** Mnemonic / trick — jaldi yaad karne ke liye.

📝 **Paper mein:** Board mein kaise poocha jata hai aur poore marks kaise.

⚠ **Dhoka:** Aam ghalti jis se marks katte hain.

★ **Golden** = sab se zyada aane wale 32 topics — pehle yeh. Har din ke aakhir mein **closed-book drill** + jawab.

Naam: _____ Class / Section: _____ Roll no.: _____

Book
XII

Fehrist — Table of Contents

33 cheat-sheet pages · 6 chapters · Golden blitz · Mock paper & key

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30-din progress tracker — har din ke baad bharo

Day	Date	Drill	Done
Day 01		/ 4	☐
Day 02		/ 4	☐
Day 03		/ 4	☐
Day 04		/ 4	☐
Day 05		/ 4	☐
Day 06		/ 4	☐
Day 07		/ 4	☐
Day 08		/ 4	☐
Day 09		/ 4	☐
Day 10		/ 4	☐

Day	Date	Drill	Done
Day 11		/ 4	☐
Day 12		/ 4	☐
Day 13		/ 4	☐
Day 14		/ 4	☐
Day 15		/ 4	☐
Day 16		/ 4	☐
Day 17		/ 4	☐
Day 18		/ 4	☐
Day 19		/ 4	☐
Day 20		/ 4	☐

Day	Date	Drill	Done
Day 21		/ 4	☐
Day 22		/ 4	☐
Day 23		/ 4	☐
Day 24		/ 4	☐
Day 25		/ 4	☐
Day 26		/ 4	☐
Day 27		/ 4	☐
Day 28		/ 4	☐
Day 29		/ 4	☐
Day 30		/ 4	☐

Rozana 3 ghante

0–15′ kal ka ★ recap · 15–105′ aaj ke cards · 105–135′ closed-book drill · 135–180′ ek 5-mark answer.

Print tips

Chrome → Print → A4 · margins **None** · **Background graphics** ON · double-sided theek hai. Mock (Day 28) ke baad wale answer pages alag rakho.

Day 01

HCI Fundamentals, Interaction & Applications

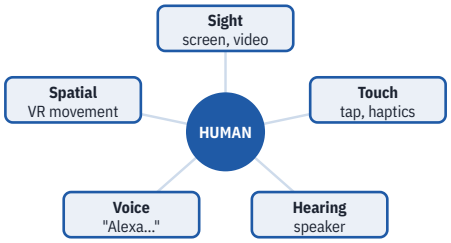
Fundamentals of HCI · Traditional & Natural Interaction · Applications (Domains) · Components

★ **Must lock:** Applications of HCI (Domains of Life)

1.1 Fundamentals of HCI

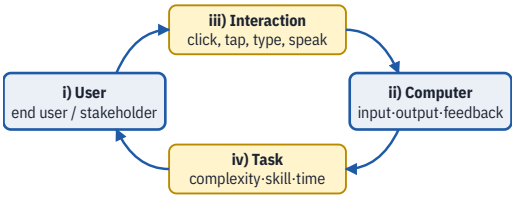
Def (paper mein yehi likho): *Human-Computer Interaction (HCI) is the study and design of technology so that it serves people.*

Achha HCI system = **easy to use, efficient**, insani zaroorat pe **responsive**. Doosre naam: **CHI** (Computer-Human Interface), **MMI** (Man-Machine Interaction) — teeno ka matlab: insaan aur machine ka milna.



🧠 **Yaad rakho:** HCI = CHI = MMI → "Human meets Machine". Paanch channels **S-T-H-V-S**: Sight, Touch, Hearing, Voice, Spatial.

1.3 Components of HCI

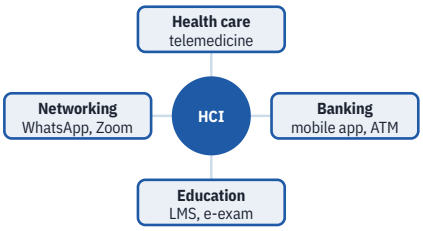


Component	Matlab
i) User	Jo bhi system se goal ke liye interact kare: **end users** (khud chalate, beginner se expert) + **stakeholders** (nateeje mein dilchaspi: managers, clients)
ii) Computer system	Platform jahan input, output, feedback interface ke zariye
iii) Interaction	User ke actions (click, tap, type, bolna) → user ka iraada system response mein
iv) Task	Khaas kaam (ticket booking): **complexity** (form vs 3D model), **user knowledge**, **time** (ATM vs passport)

🧠 **Yaad rakho: U-C-I-T**: "U Can Interact with Tasks." Task ke 3 sawal: kitna mushkil? kaunsi skill? kitna waqt?

1.2 Applications of HCI (Domains of Life) ★ Golden

Def (paper mein yehi likho): *HCI is used to design and develop interactive systems (desktop apps, websites, mobile apps) as human-centred solutions across many domains of life.*



Domain	Application examples	User benefit
Health care	Patient portals, telemedicine apps, assistive tech	Door se medical services tak rasai
Banking	Mobile banking apps, secure ATM interfaces	Tez transactions, mehfooz financial management
Education	E-learning (LMS), online exam portals	Behtar teaching, learning, engagement
Networking & communication	Social media, WhatsApp, Zoom	Duniya bhar ke log jure, collaboration

🧠 **Yaad rakho: H-BEN**: Health, Banking, Education, Networking — har ek ke liye 1 application + 1 user benefit.

📝 **Paper mein:** 5–8 marks: "Explain applications of HCI in different domains with examples." Chaar-row table + hub diagram.

1.1.1–1.1.2 Traditional vs Natural Interaction

Feature	Traditional	Natural
Main input	Mouse, keyboard, keypad, saada tap/swipe	Awaaz, chehra, haath/body gesture, harkat
Kaisa lage	Machine chalana	Insaan se baat/harkat jaisa
Examples	TV remote, ATM keypad, PC pe school-portal login	FaceID, Siri, Kinect, "Hey Google"
Output	Screen, speakers, vibration	Bola hua jawab, unlock, VR body tracking

🧠 **Yaad rakho:** Test: keys/labelled buttons dabane parein → **traditional**. Machine awaaz, chehra ya body samjhe → **natural**.

Drill: (1) HCI ke do aur naam? (2) ATM keypad traditional ya natural? (3) Chaar domains? (4) End user vs stakeholder?

Jawab: 1. CHI, MMI · 2. Traditional · 3. Health care, banking, education, networking & communication · 4. End user khud chalata; stakeholder ko nateeje se dilchaspi (manager, client) | **5-mark:** Applications of HCI.

Day 02

Interaction Types, Interfaces, Importance & Accessibility

Types of User Interaction · Interface, Environment, Feedback · Importance of HCI · Accessibility

★ **Must lock:** Importance of HCI

Types of User Interaction

Technology ke hisaab se interaction kai tareeqon se. Board aksar "list & explain 5+" poochta hai — har ek ki ek saaf misaal:

Group	Type	Misaal
Screen pe haath	Direct manipulation	File drag karke folder mein
	Gesture-based	Pinch to zoom, VR mein haath hilana
	Haptic	Phone ka vibration feedback
Alfaaz	Command-based	Linux: `cd ..`
	Conversational	Bank chatbot se chat
	Voice	"Alexa, play a song"
Lists & blanks	Menu-based	ATM menu se option chuno
	Form-based	Admission form bharna
Automatic / body	Context-aware	Raat ko phone khud dark mode

🧠 **Yaad rakho:** Groups: haath (direct, gesture, haptic) · alfaaz (command, conversational, voice) · lists/blanks (menu, form) · automatic/body (context-aware, natural).

Interface Types, Environment & Feedback

Components v–vii: **v) Interface** (controls ki shakal), **vi) Environment** (jagah jahan use ho — shor, roshni, safar), **vii) Feedback** (signal "ho gaya" / "dobara koshish").



Interface	Kaise	Misaal
GUI	Windows, icons, menus	Windows OS, Android
CLI	Text type karna	MS-DOS, Linux terminal
Touch	Tap, swipe, pinch	Smartphone, tablet
VUI	Bole hue commands	Smart speakers
NUI	Gesture, chehra, body	Kinect, FaceID

🧠 **Yaad rakho:** Feedback rule: user ke har action ka system **foran** jawab de — warna user dobara click karta hai ya ghabra jata hai.

1.4 Importance of HCI ★ Golden

Def (paper mein yehi likho): *HCI ensures that machines are approachable and usable, turning complex technology into everyday tools.*

Achha design → aasaan aur mazedaar (touchscreens). **Bura design** → uljhan aur khatra (medical device pe ghair-wazeh buttons). Is liye HCI sajawat nahi, zaroorat hai.

Implication	Matlab
1. Enhance **P**roductivity	Tez kaam, kam ghaltiyan, goals jaldi
2. Promotes **I**nclusivity	Screen readers, voice commands — har kisi ke liye barabar istemaal
3. Drives **I**nnovation	AR/VR, AI, IoT tabhi apnaye jayenge jab HCI mazboot ho
4. **E**conomic impact	User-friendly product → zyada adoption, loyalty, munafa
5. **E**thical responsibility	Privacy ka ehtiraam, bias kam, insani behbood

🧠 **Yaad rakho:** **P-I-I-E-E**: "Please Include Innovative, Ethical Earnings."

📝 **Paper mein:** 8 marks: "Why is HCI important? Discuss its implications." Opening (achha vs bura design) + 5 numbered points, har ek ki misaal.

1.5 Accessibility Principles

Def (paper mein yehi likho): *Accessibility principles are HCI guidelines that ensure digital systems are usable by all individuals, including people with diverse physical, sensory and cognitive abilities.*

Principle	Matlab	Misaal
Contrast	Text/tasveer background se wazeh	Railway app: safed pe kala text, dhoop mein bhi
Fonts	Saaf, saada font, sahih size/spacing	School site pe bara Calibri
Colour-blind friendly	Sirf rang pe bharosa nahi — symbol/label bhi	Laal circle + "STOP" label
Navigation	Ghoomne ke kai tareeqe	Mouse na ho to **Tab** key se buttons
Multimodal	Kai formats (dekhna, sunna, choona)	Video captions, phone vibration

🧠 **Yaad rakho:** **C-F-C-N-M**: "Can Friends Catch New Messages?"

Drill: (1) Linux `cd ..` kaunsi type? (2) Paanch UI types? (3) Bure design ki misaal? (4) Sirf laal "STOP" kyun masla?

Jawab: 1. Command-based · 2. GUI, CLI, Touch, VUI, NUI · 3. Medical devices pe ghair-wazeh buttons · 4. Colour-blind user ko matlab samajh nahi aata agar symbol/text na ho | **5-mark:** Importance of HCI.

Need Analysis, HCI Problems, Fixes & UI vs UX

1.6 Need Analysis for Interface Design

Def (paper mein yehi likho): *Need analysis is the process of identifying the gap between the existing situation and the desired outcome.*

1. Users pehchano → 2. Info collect → 3. Needs analyse → 4. Prioritise → 5. Solutions

UI design se **pehle** — users ke goals, tasks, salahiyat, tawaqqoat samajhne ke liye. Chhora to andaza; isi tarah school portal mein 40 menus aur "Download date sheet" ka button ghaib.

🧠 **Yaad rakho:** Need analysis = **gap** dhoondna. Existing = abhi kya takleef. Desired = "ho gaya" kaisa dikhe. UI woh gap band kare, sajaye nahi.

1.7 HCI Problems ★ Golden

Def (paper mein yehi likho): *HCI problems are common difficulties that stop users from using technology smoothly; they happen when systems fail to meet human needs or cognitive limits.*

Problem	Wazahat + misaal
Interface complexity	Bohat options; shopping app mein checkout button na mile
Poor feedback	Pata na chale kaam hua ya nahi; Submit pe screen freeze, koi confirmation nahi
Inconsistency	Ek page pe 🔍, doosre pe 🔭 search icon
Accessibility gaps	Video tutorial bina captions (behre users)
Compatibility issues	Laptop pe chale, mobile browser pe crash
Contextual neglect	Map app mein dark mode nahi — raat ko driver ki aankhein chundhiya jayein
High cognitive load	Ek saath teen SMS codes yaad rakhna
Trust deficiency	Card maange par SSL/HTTPS 🔒 nahi

🧠 **Yaad rakho:** Do families: **screen problems** (complexity, inconsistency, accessibility, compatibility, context) + **mind & trust** (feedback, cognitive load, trust).

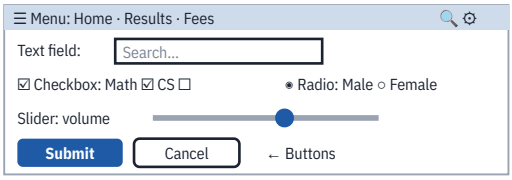
📝 **Paper mein:** 8 marks: "Describe problems of HCI with examples" — kam az kam 6: Naam → ek line → misaal.

1.8 Methods for Improving HCI

Problem (1.7)	Method (1.8)
"Send Money" nahi mil raha	**Usability** — ek wazeh button
Help video bina captions	**Accessibility** mazboot karo
Search icon har page pe alag	**Consistency** — same "Save" button har jagah
Submit freeze, koi message nahi	**Clear feedback** — "Payment Successful"
Portal mobile pe crash	**Adaptability** — responsive layout
Aise features jo kisi ne na maange	**UCD** (User-Centred Design) — interview, phir pilot-test
Buzurg lamba number type na kar sake	**Multimodal** — voice + touch
Launch ke baad shikayatein ignore	**Continuous testing** & updates

1.9 UI Design (UI vs UX) ★ Golden

Def (paper mein yehi likho): *UI design is the process of creating and organizing the visual elements of a software application so users can interact with it easily and efficiently.*



Component	Kaam
Buttons	Click se action (Submit, Cancel)
Menus	Options ki list — navigation
Icons	Tools/actions ke chhote symbols
Text fields	Type karne ke dabbe (password, search)
Checkbox / Radio	Checkbox = kai choices; radio = group mein **sirf ek**
Sliders	Bar pe value (volume, brightness)

UI (User Interface)	UX (User Experience)
Jo **dikhta** hai — rang, buttons, icons, layout	Jo **mehsoos** hota hai — poora safar, aasaani, satisfaction
Tool: Figma (visual design)	Research, interviews, testing
Misaal: JazzCash ka khoobsurat "Send" button	Kya paisa 3 taps mein bina ghabrahat ke chala gaya?

🧠 **Yaad rakho:** UI = **kapre** (outfit). UX = **poora din** (waqt pe pohanche? bheege? pareshaan hue?).

📝 **Paper mein:** Sab se zyada aane wala short Q: "Differentiate UI and UX with an example" — dono definitions + "UI is what you see; UX is how it feels" + misaal.

Drill: (1) Need analysis kab? (2) Payment page pe HTTPS lock nahi — kaunsa problem? (3) UCD full form + extra step? (4) Chhe UI components?

Jawab: 1. UI design se pehle · 2. Trust deficiency · 3. User-Centred Design; asal user goals + pilot-test · 4. Buttons, menus, icons, text fields, checkboxes/radio, sliders | **5-mark:** HCI Problems / UI vs UX.

Day 04

Wireframes, Prototypes, Evaluation & Testing

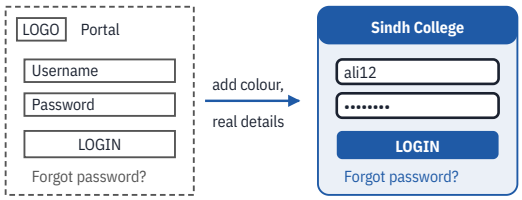
Wireframing & Figma · Test & Evaluate HCI · Testing Methods · Careers in UI/UX

★ **Must lock:** Testing Methods (Usability, A/B, Automated)

1.9.3–1.9.5 Wireframing, Figma, Prototypes

Def (paper mein yehi likho): *Wireframing is creating a basic visual outline or blueprint of a user interface before the final design.*

Lo-Fi sketch → Details add → Hi-Fi prototype



School-portal login: left Lo-Fi (dabbe + labels), right Hi-Fi (rangeen, asli jaisa)

Stage	Kya
Lo-Fi wireframe	Black & white sketch — sirf layout (boxes, lines); jaldi ideas test/share
Hi-Fi design	Rang, fonts, asli images — taqreeban final app
Prototype	**Clickable** model — screens jure hue

Figma teeno stages rakh sakta hai. Prototype linking: **Trigger** = user kya kare (On click) → **Action** = fake app kahan jaye (Navigate to "Dashboard").

🧠 **Yaad rakho:** Lo-Fi = dabbe. Hi-Fi = taqreeban asli app. Prototype = click ho sake.

1.11 Test & Evaluate HCI

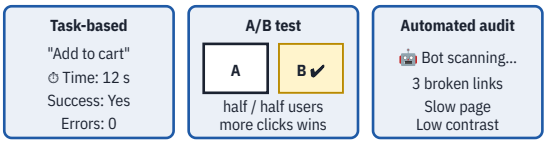
Testing = product theek chalta hai? (bugs). **Evaluation** = users ki zaroorat poori karta hai? (usability). Portal "bug-free" ho kar bhi bekaar agar date sheet na mile.

Method	Matlab
SUS (System Usability Scale)	Standard **10-item** questionnaire → usability score
NPS (Net Promoter Score)	"Dost ko recommend karoge?" → loyalty
Formative evaluation	Development ke **dauran** — jaldi feedback
Summative evaluation	Development ke **aakhir** mein — goals ke muqable kamyabi
Expert reviews	Specialists design standards (**heuristics**) se check

🧠 **Yaad rakho:** **Form**ative = bana rahe hain (dauran). **Sum**mative = jama-tafreeq (aakhir).

1.11.2 Testing Methods ★ Golden

Def (paper mein yehi likho): *Testing involves observing how the system behaves under specific conditions.*



Method	Kaise
Usability testing	Asli users ko asli kaam karte dekho — efficiency & satisfaction; user phanse to design ka masla
Task-based testing	Khaas kaam ("product dhoondo, cart mein dalo") — **time, success rate, errors** naapo
Automated audits	Software bots: broken links, slow loading, low colour contrast
A/B testing	Do versions; aadhe users A, aadhe B; zyada clicks/sign-ups wala jeete
Load testing	Ek saath bohat users (result day) — system tike ga?

A/B misaal: "Register" button sabz (A) vs neela (B) — 1000 users, B pe 12% zyada sign-ups → B rakho.

🧠 **Yaad rakho:** Task-based ke 3 numbers: **Time, Success, Errors**. A/B: do versions, users baanto, zyada clicks jeete. Bot: links, speed, contrast.

📝 **Paper mein:** Golden 8 marks: "Explain testing methods used in HCI" — table bana ke har method: definition + kya naapta + misaal.

Careers in UI/UX

Career	Kaam
UX designer	User needs research — app intuitive aur satisfying
UI designer	Visual layout (rang, typography, buttons) — Figma
Interaction designer	Screens ka flow, animations, gestures
Usability researcher	SUS/NPS aur A/B tests asli users ke saath
Accessibility specialist	Disabled logon ke liye inclusive standards

🧠 **Yaad rakho:** UI dikhata; UX poochta; interaction jorta; research naapta; accessibility sab ko shamil karta.

Chapter 1 Exam Radar 🎯

- Applications of HCI (H-BEN table) — 5–8 marks
- Importance of HCI (P-I-I-E-E) — 8 marks
- HCI problems (6 of 8 with examples) — 8 marks
- UI vs UX with example — short
- Testing methods (usability, task, A/B, audit) — 8 marks

Checkpoint: (1) Lo-Fi wireframe kya jaan-boojh ke chhorta hai? (2) Testing vs evaluation? (3) Task-based ke teen measurements? (4) SUS, NPS, A/B kaun chalata hai?

Jawab: 1. Rang aur detailed graphics · 2. Testing = bugs/chalta hai; evaluation = user needs (usability) · 3. Completion time, success rate, errors · 4. Usability researcher | 60% se kam → kal Golden repair.

Day 05

Correctness: Trace Tables & Stepwise Reasoning

Correctness · Trace Tables · Stepwise Reasoning · Comparison · Clarity

★ **Must lock:** Trace Tables, Comparing Trace & Stepwise

2.1 Analyse Algorithms for Correctness

Def (paper mein yehi likho): *An algorithm is correct if its output matches the expected results for all valid inputs.*

Ek "khush-qismat" misaal kaafi nahi — **har** valid input pe sahih ho. Do tareeqe: **trace tables** (kaghaz pe line-by-line) aur **stepwise reasoning** (logic blocks, overall state).

⚠ **Dhoka:** "Chal gaya" ≠ "sahih hai". Logical error bhi output deta hai — bas **ghalat** output.

2.1.1 Trace Tables ★ Golden

Def (paper mein yehi likho): *A trace table is a technique used to monitor how the values of variables change during the execution of an algorithm.*

Kaam: **dry running** (haath se line-by-line), **debugging** (logical errors), **validation** (wahi kare jo chahiye). Columns: line/step · variables · condition check · input/output.

```
AllHeights = [1.91, 2.01, 1.87, 1.57, 1.48]
Total = 0
for height in AllHeights:
    if height >= 1.57 and height <= 1.91:
        Total = Total + 1
print(Total)
```

Step	height	1.57 ≤ h ≤ 1.91?	Total	Output
0	—	—	0	
1	1.91	True	1	
2	2.01	False	1	
3	1.87	True	2	
4	1.57	True	3	
5	1.48	False	3	
end			3	**3**

🧠 **Yaad rakho:** Har baar variable badle ya condition test ho → **nayi row**. Na badalne wali values neeche copy karo.

📝 **Paper mein:** "Dry-run using a trace table and state the output" — har variable ka column + condition column + aakhir mein output.

2.1.2 Stepwise Reasoning (Logic Verification)

Algorithm ko logical hisson mein tod ke har stage pe **state** check: sahih shuru? sab cases? conditions theek?

```
N = 5
Sum = 0
for i in range(1, N + 1):
    Sum = Sum + i
print(Sum)
```

Check	Kya verify
Initial state	Loop se pehle Sum = 0 ✔
Loop boundary	i = 1 se N tak — na kam, na zyada
Iteration	k iterations ke baad Sum = 1+2+...+k
Final state	Sum = 1+...+N = jo chahiye
Edge N = 0	Loop nahi chalta, Sum = 0 ✔
Check N = 5	15 = $\frac{N(N+1)}{2} = \frac{5\times6}{2}$ ✔

🧠 **Yaad rakho:** Stepwise = teacher proof check kare ("ye block hamesha sahih?"). Trace = cashier sikke gine.

2.1.3 Trace Table vs Stepwise Reasoning ★ Golden



Feature	Trace table	Stepwise reasoning
Best for	Chhote blocks, complex loops, beginners	Bare systems, architecture, math proofs
Effort	High — 1000 dafa loop pe na-mumkin	Moderate — mazboot logic chahiye
Goal	Line-by-line arithmetic/misread errors	Design ki conceptual/structural ghaltiyan
Level	**Micro** — exact values	**Macro** — overall state

📝 **Paper mein:** Takreeban guaranteed: "Compare a trace table with stepwise reasoning" — 4-row table (best for, effort, goal, level).

2.2 Clarity of an Algorithm

Def (paper mein yehi likho): *Clarity means an algorithm is easy to read, understand and modify.*

Aksar thori speed se zyada ahem — maintain, debug, extend aasaan. Do pehlu: **Modularity** (hisse) aur **Readability** (text aur flow).

Drill: (1) Algorithm kab correct? (2) Trace table ke 4 columns? (3) Stepwise reasoning ka doosra naam? (4) Micro-level kaun? (5) Clarity?

Jawab: 1. Har valid input pe expected output · 2. Line/step, variables, input/output, condition check · 3. Logic verification · 4. Trace table · 5. Parhna, samajhna, badalna aasaan | **5-mark:** Trace Tables.

Modularity, Readability & Big O

Modularity · Readability · Efficiency · Number of Steps (Big O) · Number of Conditions

★ **Must lock:** Number of Steps (Big O Notation)

2.2.1 Modularity

Def (paper mein yehi likho): *Modularity means breaking an algorithm into smaller, independent parts such as functions, classes or modules.*

Idea	Matlab
SRP (Single Responsibility)	Har module **ek** kaam. Sum + report + print ek function mein = ghalat
Loose coupling	Ek hissa badlo to doosra na toote; modules reuse ho sakein
Abstraction	Andar ki tafseel chhupao — sort() call karo bina jaane kaunsa algorithm

🧠 **Yaad rakho:** SRP: "one function, one job". Naam chhota verb-phrase na ban sake (`CalculateSum`) → zyada kaam kar raha hai.

2.2.2 Readability

Def (paper mein yehi likho): *Readability is how easily another person can understand the purpose and logic of an algorithm without running it.*

Pehlu	Matlab
Naming conventions	Bura: `v = d / t` → Achha: `velocity = distance / time`
Self-documentation	Code khud samjhaye — kam comments chahiye
Simple control flow	IF ke andar IF ke andar IF se bacho
Consistency	Ek style: indentation, spacing, camelCase ya snake_case

```python
# Hard to read
v = d / t
if v > s:
    if t > 0:
        print(v)
```

```python
# Readable
velocity = dist / time
if velocity > limit:
    print(velocity)
```

🧠 **Yaad rakho:** **N-S-S-C**: Names, Self-documentation, Simple flow, Consistency.

2.3 Assess Algorithm Efficiency

Def (paper mein yehi likho): *Algorithm efficiency is measured in terms of time (speed) and space (memory).*

Teen cheezein gino: **steps** (Big O), **conditions** (IF), **repetitions** (loops). Hamesha "jab input **bare**". 10 items pe foran, 10 million pe freeze ho sakta.

2.3.1 Number of Steps (Big O) ★ Golden

Def (paper mein yehi likho): *Big O notation is a mathematical method used to describe how the running time (or memory) grows as the input size n grows.*

Chhote constants chhoro, sab se bara term rakho:

$$3n + 5 \Rightarrow O(n)$$

;

$$2n^2 + 7n \Rightarrow O(n^2)$$

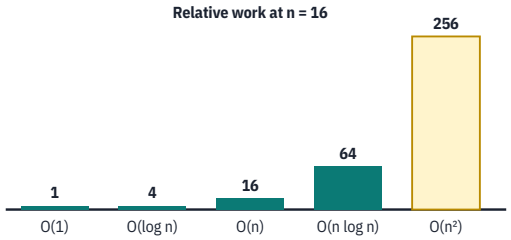
.

$$\log n$$

=

$$\log_2 n$$

(n ko kitni baar aadha karo ke 1 bache).



Class	Naam	n barhe to	Misaal	Grade
$O(1)$	Constant	Waqt same	`arr[0]`	Excellent
$O(\log n)$	Logarithmic	Bohat aahista; har step aadha	Binary search	Great
$O(n)$	Linear	n dugna → waqt dugna	Linear search	Fair/Good
$O(n \log n)$	Linearithmic	Linear se thora bura	Merge Sort	Good
$O(n^2)$	Quadratic	n² — nested loops	Bubble Sort	Poor

Tez se aahista:

$$O(1) < O(\log n) < O(n) < O(n \log n) < O(n^2)$$

. n = 16 pe: 1, 4, 16, 64, 256.

🧠 **Yaad rakho:** "**C**onstant **L**ogs **N**ever **N**eed **S**quares" — Constant, Log, n, n log n, Square.

📝 **Paper mein:** MCQ: "arr[0] access?" →

$$O(1)$$

. "Binary search?" →

$$O(\log n)$$

. "Nested loops?" →

$$O(n^2)$$

. Paanchon rows (naam, growth, misaal, grade) yaad.

2.3.2 Number of Conditions

Har IF ek **decision point** — CPU ko True/False nikalna parta = extra kaam. Nested IFs = brute force, readability aur speed dono kharab.

Short-circuiting: nateeja maloom hote hi ruk jao. AND mein pehla False → doosra check hi nahi.

```python
x, y = 0, 10
# x != 0 False → divide hota hi nahi
if x != 0 and (y / x) > 2:
    print("ratio is large")
else:
    print("skipped safely")
```

🧠 **Yaad rakho:** AND mein sasta / zyada-tar-False test **pehle**; OR mein zyada-tar-True test pehle.

Repetitions, Framework, Data Structures & Array

Number of Repetitions · Evaluation Framework · Refinements · Data Structure · Array

★ **Must lock:** Concept of Data Structure

2.3.3 Number of Repetitions

Loops efficiency ka **sab se bara** factor:

Repetition	Asar	Grade
Linear loop	Har element ek dafa	Good — $O(n)$
Logarithmic	Har iteration problem aadha (binary search)	Excellent — $O(\log n)$
Nested loops	Har element ke liye baqi sab	Poor — $O(n^2)$

🧠 **Yaad rakho:** Sab se gehra loop dhoondo: ek loop ≈ $O(n)$; loop ke andar loop ≈ $O(n^2)$; aadha karne wala ≈ $O(\log n)$.

2.3.4 Efficiency Evaluation Framework

Scale karta? → Bottleneck kahan? → Extra complexity worth it?

n 10× ho to waqt 10× (linear) ya 100× (quadratic, 10^2)?

Situation	Faisla	Kyun
50 unsorted class names	**Linear search**	n chhota; binary ke liye pehle sort = extra kaam/code
1,000,000 sorted account numbers, din bhar search	**Binary search**	Bara n; search loop bottleneck; binary standard

🧠 **Yaad rakho:** **S-B-W**: Scale, Bottleneck, Worth it.

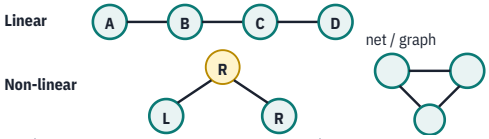
2.4 Refinements

Clarity: N-M-I-C	Efficiency: S-A-C-D
Names: `average = totalSum / totalItems`	**S**hrink steps (faltu kaam hatao)
Modules: `CalculateSum()` alag	Behtar **A**lgorithm chuno
Indentation & spacing	**C**ache — dobara hisaab mat karo
Comments — "kyun", sirf "kya" nahi	Sahih **D**ata structure

2.5 Concept of Data Structure ★ Golden

Def (paper mein yehi likho): *A data structure is a systematic way of organising, storing and managing data so that an algorithm can access, modify and process it efficiently.*

Shelf pe tarteeb se kitaabein, folders mein files — memory mein yahi kaam data structures karte hain.



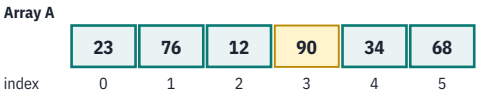
	Linear	Non-linear
Tarteeb	Ek ke baad ek (sequence)	Shaakhein / jaal
Rishta	Har item ka 1 pichla + 1 agla	Ek item ke kai connections
Misaal	Array, linked list, stack, queue	Tree, graph

🧠 **Yaad rakho:** "Har item ka zyada se zyada **ek** agla?" Haan → linear. "Ek se kai, ya jaal?" → non-linear.

📝 **Paper mein:** Poori definition (organise, store, manage; efficient access) + linear vs non-linear ("one predecessor, one successor") + misaalein.

2.5.1 Array

Def (paper mein yehi likho): *An array holds a collection of similar items (same data type) in contiguous memory; each item has an index starting from 0.*



Contiguous = ek musalsal block. Index se address nikal aata → seedha jump: **direct (random) access** = $O(1)$. `A[3]` → 90 bina 23, 76, 12 se guzre.

```
A = [23, 76, 12, 90, 34, 68]
print(A[3])   # 90
print(A[0])   # 23
```

🧠 **Yaad rakho:** Index = baayen se position − 1. Insaan ka "4th" = index 3.

5-mark frame: Concept of Data Structure

Define	A systematic way of organising, storing and managing data so an algorithm can access, modify and process it efficiently.
Explain	Linear: sequence, ek agla. Non-linear: shaakhein/jaal.
Example	Array, linked list, stack, queue · tree, graph.
Diagram	A–B–C–D line + root wala tree.
Working	Array: A[3] direct jump O(1).

Drill: (1) Nested loops ka grade? (2) n 10× → quadratic kitna slow? (3) `a = b / c` average ke liye? (4) Data structure define karo (5) Contiguous memory?

Jawab: 1. Poor — O(n²) · 2. ~100× · 3. `average = totalSum / totalItems` · 4. Organise, store, manage for efficient access · 5. Items ek musalsal block mein saath saath | **5-mark:** Concept of Data Structure.

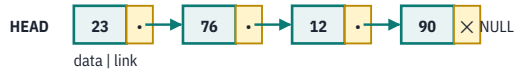
Linked List, Stack, Queue, Tree & Graph

Linked List · Stack (LIFO) · Queue (FIFO) · Tree · Graph

★ **Must lock:** Stack, Queue

Linked List

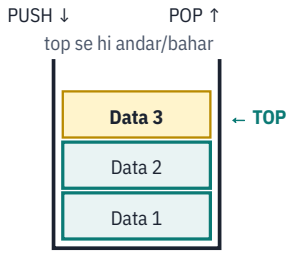
Data ki **zanjeer** — array ki tarah ek block nahi. Har **node** = **data** + **link (pointer)** jo agle node ka address rakhe. Aakhri link = **NULL**.



🧠 **Yaad rakho:** Array = number wale lockers ki qataar. Linked list = khazane ki talaash: har clue agle dabbe ka pata.

Stack (LIFO) ★ Golden

Def (paper mein yehi likho): *A stack is a linear data structure that works on the LIFO principle: Last-In, First-Out.*



PUSH = top pe add. **POP** = top se nikalo. **TOP** = sab se upar wala.

Misaal: Undo/Redo, Back button, function calls.

#	Operation	Stack (bottom → top)	TOP	Popped
1	PUSH Data1	[Data1]	Data1	
2	PUSH Data2	[Data1, Data2]	Data2	
3	PUSH Data3	[Data1, Data2, Data3]	Data3	
4	POP	[Data1, Data2]	Data2	**Data3**
5	POP	[Data1]	Data1	**Data2**

```
stack = []
stack.append("Data1")   # PUSH
stack.append("Data2"); stack.append("Data3")
print(stack.pop())      # POP → Data3
```

🧠 **Yaad rakho:** Canteen ki plates: aakhir mein dhuli plate pehle use. **LIFO**.

📝 **Paper mein:** Stack + LIFO, PUSH/POP, TOP, Undo misaal, vertical boxes TOP label ke saath. "Show the stack after PUSH…POP".

Queue (FIFO) ★ Golden

Def (paper mein yehi likho): *A queue organises data on the FIFO principle: First-In, First-Out.*



ENQUEUE = **Rear** (peeche) pe add. **DEQUEUE** = **Front** se nikalo. Misaal: printer jobs, canteen ki line.

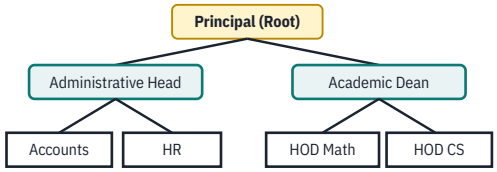
#	Operation	Queue (Front → Rear)	Front	Rear
1	ENQUEUE 55	[55]	55	55
2	ENQUEUE 76	[55, 76]	55	76
3	ENQUEUE 90	[55, 76, 90]	55	90
4	ENQUEUE D	[55, 76, 90, D]	55	D
5	DEQUEUE → **55**	[76, 90, D]	76	D

🧠 **Yaad rakho:** **FIFO** = canteen line: Front = jise serve ho raha; Rear = jo abhi aaya.

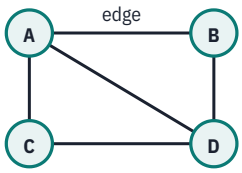
📝 **Paper mein:** Queue + FIFO, ENQUEUE (rear) / DEQUEUE (front), printer misaal, Front/Rear diagram, chhota trace.

2.5.2 Tree (non-linear)

Hierarchy — sab ek **root** se phootta; **one-to-many** (parent–child). Koi child do parents ka nahi.



Graph (non-linear)



Nodes (vertices) = cheezein (shehar, log). **Edges** = connections (sarak, dosti). Cycle ban sakta (A–B–D–C–A) — tree mein nahi.

🧠 **Yaad rakho:** Tree = khas graph: ek root, koi cycle nahi. Cycles / many-to-many → graph.

Array vs Linked List

Point	Array	Linked list
Memory	Contiguous (ek block)	Bikhri nodes, links se jurri
Access	Direct by index $O(1)$	Sequential, HEAD se $O(n)$
Insert / delete	Beech mein mushkil (shift)	Aasaan — link badlo
Size	Aam tor pe fixed	Dynamic, barh/ghat sakta

Stack vs Queue

Point	Stack	Queue
Principle	LIFO	FIFO
Add	PUSH — top pe	ENQUEUE — rear pe
Remove	POP — top se	DEQUEUE — front se
Ends used	Ek (top)	Do (front, rear)
Misaal	Undo, back button	Printer, canteen line

Drill: (1) Node ke 2 hisse? (2) LIFO? (3) FIFO — add kahan, remove kahan? (4) Tree ka starting node? (5) Graph ke 2 hisse?

Jawab: 1. Data + link · 2. Last-In First-Out · 3. First-In First-Out; Rear pe add, Front se remove · 4. Root · 5. Nodes (vertices), edges | **5-mark:** Stack / Queue.

Operations, Choosing Structures & Data Retrieval

Operations on Data Structures · Scenarios · Trace Data Retrieval · Careers

★ **Must lock:** None — tables copy karo

2.5.3 Operations on Data Structures

Operation	Matlab	Misaal
Insertion	Naya item daalna	PUSH; ENQUEUE; array mein 40
Deletion	Item nikalna	POP; DEQUEUE; node delete
Traversal	Har item ek ek dekhna	HEAD → … → NULL; har tree node
Searching	Khaas item dhoondna	Array linear search; links follow karke 87
Sorting	Tarteeb (asc/desc)	Marks ki array pe Bubble Sort

🧠 **Yaad rakho: I-D-T-S-S**: Insert, Delete, Traverse, Search, Sort.

2.6 Scenario se Data Structure

Scenario	Structure	Kyun
Class ke grades	**Array**	Size maloom, same type, index se tez
Music playlist / browser history	**Linked list**	Har song ko agla maloom
Undo / Redo	**Stack**	Aakhri action pehle wapas
Printer jobs	**Queue**	Pehle aao, pehle pao
College organogram, folders	**Tree**	Parent–child
Maps, social network	**Graph**	Connections / rishte

🧠 **Yaad rakho:** Ulti tarteeb? → stack · Pehle aao? → queue · Parent–child? → tree · Many-to-many? → graph · Index wali list? → array.

Practice trace (khud karo)

Stack: PUSH 5, PUSH 8, POP, PUSH 3, PUSH 9, POP, POP → bacha? popped tarteeb?

Queue: ENQUEUE A, B, C, DEQUEUE, ENQUEUE D, DEQUEUE → Front? Rear?

Jawab: stack [5], popped 8, 9, 3 · queue [C, D], Front C, Rear D (A, B nikle)

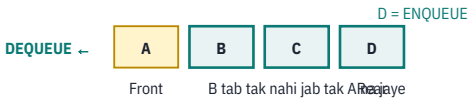
```
from collections import deque
q = deque()
q.append("A"); q.append("B")    # ENQUEUE
print(q.popleft())              # DEQUEUE → A
```

2.7 Trace Data Retrieval

Teeno linear, magar dhoondne ki mehnat alag:

Iteration	Index	Access	Value
1	0	Array[0]	10
2	1	Array[1]	50
3	2	Array[2]	20

```
Array = [10, 50, 20, 70, 90, 30]
print("Direct Array[2] =", Array[2])   # 20, O(1)
for i in range(3):
    print(i + 1, i, Array[i])
```



Structure	Retrieval
Array	**Jump** — index se direct, $O(1)$
Linked list	**Walk** — HEAD se links follow, $O(n)$
Queue	**Wait** — sirf Front se

🧠 **Yaad rakho:** "Array jumps, list walks, queue waits."

Careers in Algorithms

Career	Kaam
Software Engineer	Data structures + efficient algorithms se software
Backend Developer	Server-side logic; sahih structure → web speed
Algorithm Engineer	Search engines, finance, logistics ke algorithms optimise
Systems Analyst	Mojooda systems ki efficiency/clarity behtar
Competitive Programmer	Time-limit mein mushkil problems

🧠 **Yaad rakho:** Har career ke 3 test: **correct, clear, efficient** + sahih data structure.

Chapter 2 Exam Radar 🎯

- Dry-run with trace table + output
- Trace table vs stepwise reasoning (4-row table)
- Big O classes: naam, misaal, grade; order fastest → slowest
- Define data structure; linear vs non-linear
- Stack/queue: define, diagram, PUSH/POP · ENQUEUE/DEQUEUE trace
- Scenario → structure (undo, printer, playlist…)

Checkpoint: (1) Paanch operations? (2) Class grades kis mein, kyun? (3) [10, 50, 20, 70, 90, 30] mein Array[2]? (4) Search-engine/logistics algorithms kaun?

Jawab: 1. Insertion, deletion, traversal, searching, sorting · 2. Array — size maloom, same type, tez index · 3. 20 · 4. Algorithm Engineer | 60% se kam → kal Golden repair.

Day 10

Lists, Tuples, Sets & Dictionaries

Primitive vs Abstract · Lists · Tuples · Sets · Dictionaries

★ **Must lock:** Lists, Tuples, Dictionaries

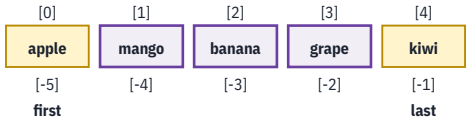
3.1 Data Structures in Python

Category	Matlab	Examples
Primitive	Sirf ek value	int, float, bool
Abstract	Bohat si values, richer operations	list, tuple, set, dict

🧠 **Yaad rakho:** Primitive = ek sikka 🪙. Abstract = batwa jis mein kai sikke.

3.1.1 Lists (Ordered, Mutable) ★ Golden

Def (paper mein yehi likho): *A list stores an ordered collection of items in a single variable; it is mutable and written with square brackets [ ].*



Method / op	Kaam
`.append(x)`	Aakhir mein add
`.insert(i, x)`	Index i pe add
`.remove(x)`	Pehla x hatao
`.pop(i)`	Index i (default aakhri) nikal ke wapas do
`.sort()` / `.reverse()`	Tarteeb / ulta (original badle)
`.index(x)` / `.count(x)`	Pehli position / kitni dafa
`L[2:6]`	Slice: index 2 se 5 tak

After	List
create	['apple', 'banana', 'apple']
.append('grape')	['apple', 'banana', 'apple', 'grape']
.remove('apple')	['banana', 'apple', 'grape'] — sirf pehla
.sort()	['apple', 'banana', 'grape']

🧠 **Yaad rakho:** `[-1]` hamesha aakhri item — `len(L)-1` ki zaroorat nahi. Mixed types allowed: `[1, "Python", 3.14, True]`.

📝 **Paper mein:** Har saal: "What is a list?", "4 methods", "list vs tuple", "search program", "predict output" (slicing).

3.1.2 Tuples (Ordered, Immutable) ★ Golden

Def (paper mein yehi likho): *A tuple is an ordered collection that is immutable — once created, items cannot be added, removed or replaced. Written with ( ).*

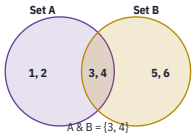
```
t = (10, 20, 30, 40, 50)
print(t[0], t[-1])    # 10 50
print(t[1:4])         # (20, 30, 40)
print(t[::-1])        # (50, 40, 30, 20, 10)
t[0] = 99             # TypeError!
```

Point	List [ ]	Tuple ( )
Mutability	Mutable — badal sakti	**Immutable** — read-only 🔒
Methods	Bohat (append, remove...)	Sirf `count`, `index`
Speed / safety	Thori slow	Tez, data mehfooz
Use	Badalta data	Fixed data (coordinates, din)

🧠 **Yaad rakho:** Tuple = band lunchbox: andar dekh sakte, sandwich badal nahi sakte. "Tuple is Tight".

3.1.3 Sets (Unordered, Unique)

Unordered, unindexed, **unique** items, `{ }`. Duplicates foran khatam: `{1,4,3,3,2,2}` → `{1,2,3,4}`.



Op	Result
`A | B`	{1..6} union
`A & B`	{3, 4}
`A - B`	{1, 2}
`A ^ B`	{1,2,5,6}

`.add()` ek · `.update()` kai · `.remove()` (na ho to **KeyError**) · `.discard()` (crash nahi) · `.clear()`

⚠ **Dhoka:** Khaali set = `set()`, **{}** nahi (woh khaali dict hai).

3.1.4 Dictionaries (Key–Value) ★ Golden

Def (paper mein yehi likho): *A dictionary stores data as key–value pairs; it is ordered (Python 3.7+), mutable, and its keys are unique.*

```
d = {"name": "Alice", "age": 30}
d["age"] = 31          # change
d["city"] = "NYC"      # add
del d["city"]          # delete
print(d.get("marks", 0))   # 0, crash nahi
```

`.keys()` · `.values()` · `.items()` (tuples). Key repeat → aakhri value: `{"a":1,"a":3}` → `{'a': 3}`. Position se nahi, **key** se dhoondo.

🧠 **Yaad rakho:** Keys = roll numbers (unique); values repeat ho sakti. "**get** is gentle, **brackets** can break" (KeyError).

Drill: (1) 2 primitive + 2 abstract? (2) `nums[2:6]` for 0..9? (3) Tuple ke 2 methods? (4) `print({1,4,3,3,2,2})`? (5) `{"a":1,"a":2,"a":3}`?

Jawab: 1. int, float · list, dict · 2. [2, 3, 4, 5] · 3. count, index · 4. {1, 2, 3, 4} · 5. {'a': 3} | **5-mark:** Lists / Dictionaries.

Built-in Functions, Activities & Functions

min/max/len/sum · Attendance System · Word Frequency · Functions · Built-in vs User-defined

★ **Must lock:** Functions in Python

3.1.5 Built-in Functions

Function	Kaam	Kis pe
`len()`	Kitne elements (dict: keys)	Sab
`min()` / `max()`	Sab se chhota / bara (numeric ya alphabetical)	List, tuple, set, dict keys
`sum()`	Numbers ka jama	List, tuple, set; dict: `sum(d.values())`
`sorted()`	Naya sorted list; original same	Sab

```
s = {5, 1, 9, 3, 7}
print(len(s), min(s))    # 5 1
print(max(s), sum(s))    # 9 25
print(sorted(s))         # [1, 3, 5, 7, 9]
```

🧠 **Yaad rakho:** LEN-MIN-MAX-SUM = gino, chhota, bara, jama. Dict pe len/min/max/sorted **keys** use karte; sum ko `.values()` chahiye.

3.1.6 Activity: Attendance System

```
students = ["Ali", "Sara", "Ayesha"]
present_count = 0
for name in students:
    ans = input(name + " present? ")
    if ans.lower() == "yes":
        present_count += 1
print("Present:", present_count)
print("Absent:", len(students) - present_count)
```

Student	Typed	.lower()	count
Ali	yes	yes → +1	1
Sara	no	—	1
Ayesha	YES	yes → +1	2

3.1.7 Activity: Word Frequency

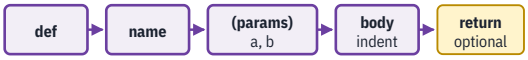
```
text = "python is easy python is fun "
text += "python is powerful"
word_freq = {}
for word in text.split():
    word_freq[word] = word_freq.get(word, 0) + 1
print(word_freq)
# {'python': 3, 'is': 3, 'easy': 1,
#  'fun': 1, 'powerful': 1}
```

🧠 **Yaad rakho:** Har counter ka dil: `d[item] = d.get(item, 0) + 1` — letters, votes, roll numbers sab.

3.2 Functions in Python ★ Golden

Def (paper mein yehi likho): *A function is a named block of code that is written once and runs whenever it is called, avoiding duplicate code.*

Same instructions bar bar = **duplicate code**: lamba, mushkil, har copy theek karni parti.



def add(a, b): → return a + b → add(2, 3) = 5

```
def greet():
    print("Hello, World!")

greet()     # call 1
greet()     # call 2
```

Benefit	Matlab
Code reusability	Ek dafa likho, har jagah use
Modularity	Bara masla chhote naam wale kaamon mein
Readability	Chhoti, saaf files, meaningful naam
Maintainability	Bug ek jagah theek → har call theek

🧠 **Yaad rakho:** **DRY** — Don't Repeat Yourself. Copy-paste ka irada → `def` mein lapeto. Naam verbs: greet, add, save_grades.

3.2.1 Built-in vs User-defined

	Built-in	User-defined
Kaun likhta	Python language	Programmer
Shuru	Bas naam call	Pehle `def name(...):`
Misaal	`len("Ali")` → 3, range, sum, print, input, type	`def add(a, b): return a + b`
Maqsad	Sab ke rozana kaam	Aap ke program ke khaas kaam

```
def greet_student(name):      # parameter
    print("Hello,", name)
greet_student("Hassan")       # argument
```

🧠 **Yaad rakho:** Python ke paas pehle se hai (len, sum, max) → dobara mat likho. Apne steps pack karne hon → user-defined.

5-mark frame: Functions

Define	A named block of code written once and run whenever it is called.
Explain	def, name, parameters, indented body, optional return.
Example	def add(a, b): return a + b → add(2, 3) = 5
Working	Benefits: reusability, modularity, readability, maintainability.

Drill: (1) s = {5,1,9,3,7}: len, min, max, sum? (2) Teeno "no" type karein to? (3) "is" kitni dafa? (4) Function kis keyword se? (5) range() built-in ya user-defined?

Jawab: 1. 5, 1, 9, 25 · 2. Present 0, Absent 3 · 3. Teen · 4. def · 5. Built-in | **5-mark:** Functions in Python.

Return Values, Scope & File Handling

Return Values · Calculator Activity · Local vs Global · File Handling & Modes r, w, a

★ **Must lock:** File Handling (Modes r, w, a)

3.2.2 Functions with Return Values

Function result **print** kare (screen pe dikhaye) ya **return** kare (caller ko wapas bheje). Returned value store, print ya aage hisaab mein use.

```
def multiply(a, b):
    return a * b
    print("never runs")      # return ke baad

result = multiply(5, 3)      # call → 15
print("The answer is:", result)
```

🧠 **Yaad rakho:** **return** = wapas bhejo; **print** = abhi dikhao. return function se **foran bahar** le jata hai.

3.2.3 Activity: Calculator

```
def add(a, b): return a + b
def subtract(a, b): return a - b
def multiply(a, b): return a * b
def divide(a, b):
    if b == 0:
        return "Cannot divide by zero"
    return a / b

x = float(input("First: "))
y = float(input("Second: "))
print(add(x, y), subtract(x, y))
print(multiply(x, y), divide(x, y))
```

🧠 **Yaad rakho:** Main program = **fehrist** (calls ka silsila), formulon ka dher nahi. divide mein zero ka guard.

3.2.4 Local vs Global Scope

Scope = program ka woh hissa jahan variable dikhe/use ho.

	Local	Global
Kahan bana	Function ke **andar**	Functions ke **bahar**
Zindagi	Sirf function chalne tak	Poora program
Kaun dekhe	Sirf woh function	Sab (badalne ke liye `global` keyword)

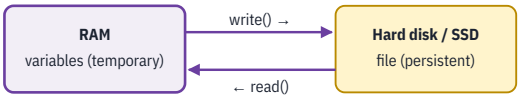
```
def calc():
    secret = 50        # local
    print(secret)
calc()
print(secret)          # NameError!
```

🧠 **Yaad rakho:** Local = kamre ke andar; global = corridor. Kamre se niklo to andar ki cheezein ghaib. Locals ko tarjeeh.

3.3–3.4 File Handling (Modes r, w, a) ★ Golden

Def (paper mein yehi likho): *File handling is a program's ability to create, read, write, update and delete files on disk.*

Variables RAM mein — program khatam, sab gaya. Files disk pe — **persistent** (rehti hain).



Type	Kya store	Misaal
Text file	Insaan parh sake (characters)	.txt, .csv, .py
Binary file	Raw bytes	.jpg, .mp3, .dat

Mode	Kaam	File na ho to	File ho to
'r' read	Parhna (default)	**FileNotFoundError**	Parhta hai
'w' write	Likhna	Nayi banata	**Mita ke** naya likhta ⚠
'a' append	Aakhir mein jorna	Nayi banata	Purana rakh ke neeche add

Method	Kaam
`read()`	Poori file ek string
`readline()`	Ek line
`readlines()`	Sab lines ki list
`write(s)`	String likho (newline khud \n)
`close()`	File band — zaroori

```
f = open("myfile.txt", "w")
f.write("The quick brown fox!\n")
f.close()

f = open("myfile.txt", "r")
print(f.read())
f.close()
```

🧠 **Yaad rakho:** **r** = Read existing · **w** = Wipe then write · **a** = Add at end. `write()` khud newline nahi dalta.

📝 **Paper mein:** Define file handling, RAM vs disk, modes r/w/a ek ek property ke saath, warning: **w overwrites**. Chhota program: ek line likho, close.

5-mark frame: File Handling

Define	A program's ability to create, read, write, update and delete files on disk.
Explain	RAM temporary, disk persistent; modes r (read), w (overwrite), a (append).
Example	grades.txt mein marks likhna aur parhna.
Diagram	RAM ⇄ Disk: write() →, ← read().
Working	open → write/read → close (ya with).

Drill: (1) `result = multiply(5, 3)` → result? (2) `divide(10, 0)`? (3) Local variable kahan? (4) Kaunsa mode overwrite karta?

Jawab: 1. 15 · 2. "Cannot divide by zero" · 3. Function ke andar · 4. 'w' | **5-mark:** File Handling (r, w, a).

with Statement, Exceptions & Grade Tracker

Context Manager · try-except · Grade Tracker Project · Careers

★ **Must lock:** The with Statement (Context Manager)

3.6 The with Statement (Context Manager) ★ Golden

Def (paper mein yehi likho): *The with statement is a context manager: it opens the file, runs the indented block, and always closes the file when the block ends — even if an error occurs.*

with open(...) as f → f use karo → block khatam → file auto-close ✔

`.close()` bhoolna aasaan; crash pe file adhuri/kharab ho sakti.

```
with open("myfile.txt", "w") as f:
    f.write("Context managers are safe!\n")
# yahan close() ki zaroorat nahi

with open("myfile.txt", "r") as f:
    for line in f:
        print(line.strip())
```

Manual open/close	with
close() khud likhna	Khud-ba-khud close
Error pe file khuli reh sakti	Error pe bhi close
Zyada lines	Saaf, chhota

🧠 **Yaad rakho: with** = "lo, use karo, wapas rakho". Mode phir bhi aap chuno: 'r', 'w', 'a'.

📝 **Paper mein:** Context manager jo close guarantee kare; manual se farq; 4-line program with + write.

3.8 Errors & Exceptions (try-except)

Exception = program chalte waqt error (missing file). Handle na karo → traceback + program ruk jata. **try/except** pakar ke saaf message dikhata.

```
try:
    with open("missing_data.txt", "r") as f:
        print(f.read())
except FileNotFoundError:
    print("Error: file does not exist.")
except Exception as e:
    print(f"Unexpected error: {{e}}")
else:
    print("Read OK")          # error na aaye to
finally:
    print("Done")             # hamesha
```

Error	Matlab
FileNotFoundError	Path maujood nahi
PermissionError	OS parhne/likhne nahi deta
IOError	Aam I/O masla (disk full/disconnect)

🧠 **Yaad rakho:** try = koshish; except = backup plan. **Specific** pehle, general `Exception` baad mein.

3.9 Mini Project: Grade Tracker

'w' names + marks → 'a' naya student → 'r' lines parho → split, total, average

```
with open("grades.txt", "w") as f:    # create
    f.write("Alice,88\nBob,75\n")
    f.write("Carol,92\n")

with open("grades.txt", "a") as f:    # add
    f.write("David,81\n")

total, count = 0, 0
with open("grades.txt", "r") as f:    # read
    for line in f:
        name, mark = line.strip().split(",")
        total += int(mark)
        count += 1
print("Average:", total / count)      # 84.0
```

$$\text{Average} = \frac{88+75+92+81}{4} = \frac{336}{4} = 84.0$$

⚠ **Dhoka:** David add karne ke liye 'w' kholi to Alice, Bob, Carol **mit** jayenge — 'a' use karo!

Careers

Career	Kaam
Python / software developer	Applications, scripts, tools
Data analyst	Lists, dicts, files se data process aur interpret
Automation engineer	Repetitive file/testing kaam scripts se
Game developer	Functions, data structures, logic se games

🧠 **Yaad rakho: P-D-A-G.**

Chapter 3 Exam Radar 🎯

- List: define, 4 methods, slicing output; list vs tuple
- Set ops (|, &, –), remove vs discard
- Dict: d[key] vs d.get(), repeated key, .items() loop
- Functions: benefits, return vs print, local vs global
- File modes r/w/a, with statement, try-except program

Checkpoint: (1) Context manager file ke liye kya guarantee karta? (2) Missing file 'r' mein → kaunsa exception? (3) Sample ka class average? (4) Lists, dicts, files se business data — kaunsa career?

Jawab: 1. Block khatam hone pe file close, error ho tab bhi · 2. FileNotFoundError · 3. 84.0 = 336/4 · 4. Data analyst | 60% se kam → kal Golden repair.

Day 14

Data Sources, SQLite, Pandas & Missing Values

Data Analysis · Database Connection · SQLite in Python · Pandas DataFrames · NaN

★ **Must lock:** Data Sources & Connection, Pandas, NaN

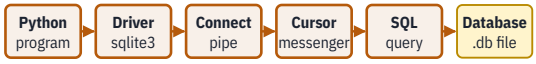
4.1 Concept of Data Analysis

Def (paper mein yehi likho): *Data analysis is examining, organising and interpreting data to extract useful information.*

School results, dukaan ki sales, hospital records — faisla, trend ya hal ke liye. Python kyun: parhne mein aasaan, files/DB se baat, statistics, graphs. Pandas ki 5 lines = ghante bhar ki spreadsheet clicking.

4.1.1–4.1.2 Data Sources & Database Connection ★ Golden

Data sources = info kahan se: school records, websites, sensors, databases. **Connectivity** = Python aur source ki baat-cheet (read/update).



#	Component	Kaam
1	Database	Organised store (SQLite = ek .db file; MySQL = bari apps ka server)
2	Driver	Python aur DBMS ka tarjuman (`sqlite3` built-in)
3	Connection object	Pipeline kholna, sambhalna, band karna
4	Cursor object	SQL chalata, result rows mein chalta
5	SQL queries	CREATE, INSERT, SELECT, UPDATE, DELETE

⚠ **Dhoka:** Cursor aur connection ulat mat karo: connection = **pipe**; cursor = **messenger** jo pipe mein safar kare.

4.2 SQLite Database in Python

Built-in; poora DB ek `.db` file — na password, na server. Steps: **connect → cursor → execute → commit → fetchall → close**.

```
import sqlite3
conn = sqlite3.connect("school.db")
cur = conn.cursor()
cur.execute("""CREATE TABLE IF NOT EXISTS
  students (id INTEGER PRIMARY KEY,
            name TEXT, marks INTEGER)""")
cur.execute("INSERT INTO students "
    "(name, marks) VALUES ('Ali', 85)")
conn.commit()                  # disk pe likho
cur.execute("SELECT * FROM students")
print(cur.fetchall())          # [(1, 'Ali', 85)]
conn.close()                   # free + unlock
```

🧠 **Yaad rakho:** SQL ke andar strings **single quotes**: 'Ali'. `SELECT *` = sab columns.

4.3 Pandas & DataFrames ★ Golden

Pandas = tables ki third-party library. Terminal mein ek dafa `pip install pandas`; convention `import pandas as pd`.

Def (paper mein yehi likho): *A DataFrame is a 2-D table of rows and columns, like Excel or an SQL table.*

```
import pandas as pd
data = {"Name": ["Ali", "Sara", "Ahmed"],
        "Math": [85, 90, 78],
        "Science": [80, 88, 70]}
df = pd.DataFrame(data)
print(df)
```

	Name	Math	Science
0	Ali	85	80
1	Sara	90	88
2	Ahmed	78	70

Command	Kaam
`pd.read_csv("f.csv")`	CSV se DataFrame
`df.head()` / `df.info()`	Pehli 5 rows / columns + types
`df.describe()`	count, mean, std, min, max...
`df["Math"].mean()`	Column ka average → 84.33
`df[df["Math"] > 80]`	Filter rows
`df.sort_values("Math")`	Sort
`df.groupby("Class").mean()`	Group-wise

4.3.3 Handling Missing Values (NaN) ★ Golden

Khaali cells = **NaN** ("Not a Number"). Nazar-andaaz karo to statistics crash ya **jhooth** bolein.

```
df = pd.DataFrame({
  "Name": ["Ali", "Sara",
           "Ahmed", "Ayesha"],
  "Marks": [85.0, None, 78.0, 92.0],
  "Grade": ["A", "A", None, "A"]})
print(df.isnull().sum())  # Marks 1, Grade 1
avg = df["Marks"].mean()
df["Marks"] = df["Marks"].fillna(avg)
df["Grade"] = df["Grade"].fillna("Unknown")
clean = df.dropna()   # NaN rows hatao
```

Tool	Kaam
`isnull()`	Jahan khaali → True
`fillna(value)`	Khaali ko value se bharo
`dropna()`	Khaali wali row delete

Sara ka Marks → mean

$$= \frac{85+78+92}{3} = 85.0$$

⚠ **Dhoka:** CEO samait company ki salary ke mean se gap bharna = ghurbat chhup jati (outlier). Pehle saaf karo, phir analyse.

Drill: (1) School mein data analysis ka ek use? (2) `SELECT *` kaun chalata? (3) * ka matlab? (4) pd? (5) `df.isnull().sum()`?

Jawab: 1. Results parh ke teaching behtar · 2. Cursor · 3. Sab columns · 4. pandas ka alias · 5. Har column mein NaN ki ginti | **5-mark:** Database connection / Pandas / NaN.

Day 15

Organization, Visualization & Descriptive Statistics

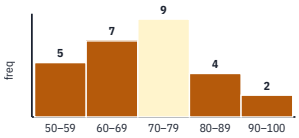
Frequency Distribution · Graph Types · Matplotlib · Mean, Median, Mode, Variance, SD

★ **Must lock:** Data Visualization, Descriptive Statistics

4.4 Data Organization & Representation

Teen roop: **Table** (exact values: Ali 85 Math, 80 Science) · **Frequency distribution** (har range/bin mein kitne) · **Graph** (trend ki tasveer).

Marks	Students
50–59	5
60–69	7
70–79	9
80–89	4
90–100	2
Total	**27**



80 ya zyada: 4 + 2 = **6** students.

4.5 Data Visualization & Graph Types ★ Golden

Sawal ke mutabiq graph chuno; **axes label + title** — bina label chart = zero marks.



Chart	Dikhata hai	Dhyan
Bar / column	Categories compare (subjects)	Column = vertical, bar = horizontal
Line	Waqt ke saath badlao	x-axis tarteeb wala (mahine)
Pie	Ek whole ke hisse (%)	Bohat / barabar slices pe bura
Histogram	Ek numeric variable ki distribution (bins)	Bars **jurri** (continuous)
Scatter	Do numeric variables ka rishta	Har dot ek (x, y)
Box plot	Min, Q1, median, Q3, max, outliers	Extreme values ke liye best

```
import matplotlib.pyplot as plt
subjects = ["Eng", "Math", "Sci", "Hist"]
marks = [64, 82, 71, 49]
plt.bar(subjects, marks, color="skyblue")
plt.title("Marks vs Subjects")
plt.xlabel("Subjects"); plt.ylabel("Marks")
plt.show()
```

📝 **Paper mein:** "Which graph and why?" — 3 marks ka tohfa: naam + kya dikhata + baqi kyun bure.

Chapter 4 Exam Radar 🎯

- 5 components of DB connection in order
- SQLite steps: connect → cursor → execute → commit → fetchall → close
- DataFrame banana; isnull / fillna / dropna
- Which graph and why · histogram vs bar
- Mean, median, mode, range, variance, SD (n vs n−1)

4.6 Descriptive Statistics ★ Golden

Lambi list ko chand imaandar numbers mein: **central tendency** ("aam kya?") aur **spread** ("kitna bikhra?").

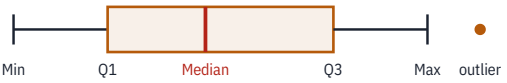
Measure	Kaise	Kab use
Mean	$\bar{x} = \frac{\sum x}{n}$ — har value	Values qareeb qareeb
Median	Sort karke beech; even n → do beech walon ka mean	Outliers ho (CEO salary)
Mode	Sab se zyada baar; koi nahi / ek / kai	Sab se popular (shoe size)
Range	Max − Min	Seedha spread

$$\sigma^2 = \frac{\sum(x-\bar{x})^2}{n} \qquad s^2 = \frac{\sum(x-\bar{x})^2}{n-1} \qquad SD = \sqrt{\text{variance}}$$

Worked: 4, 8, 6, 10, 2 → sorted **2, 4, 6, 8, 10**

x	$x - \bar{x}$	$(x-\bar{x})^2$
2	−4	16
4	−2	4
6	0	0
8	2	4
10	4	16
Σ = 30		**Σ = 40**

Mean $= \frac{30}{5} = 6$ · Median = 6 · Range = 10 − 2 = 8 · Population variance $= \frac{40}{5} = 8$, SD $= \sqrt{8} \approx 2.83$ · Sample (n−1) variance $= \frac{40}{4} = 10$, SD ≈ 3.16 (pandas `.std()` n−1 use karta).



⚠ **Dhoka:** Variance ko "marks" mein mat batao — us ki unit **square** hai; SD asal unit mein.

📝 **Paper mein:** n ya n−1 batao. Median ke liye sorted list dikhao. SD: mean → deviations → squares → divide → square root.

Careers

Career	Tools
Data analyst	Pandas, Excel
Database administrator	SQLite / MySQL
Data scientist	Statistics + modelling
Visualisation specialist	Matplotlib, Power BI

Checkpoint: (1) Table mein 80+ kitne? (2) Histogram ki bars kyun jurti? (3) 4, 8, 6, 10, 2 ka median?

Jawab: 1. 4 + 2 = 6 · 2. x-axis continuous numeric range, alag categories nahi · 3. Sorted 2, 4, 6, 8, 10 → 6 | 60% se kam → kal Golden repair.

Day 16

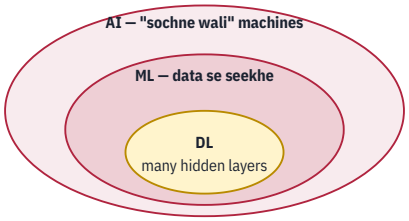
Machine Learning, Neural Networks & Deep Learning

AI ⊃ ML ⊃ DL · Machine Learning · Neural Networks · Deep Learning · NN Components

★ **Must lock:** Neural Networks, NN Components

5.1 ML, NN & DL — AI ke hisse

Misaalein: phone ka **face unlock**, speech recognition (Siri, Alexa, Urdu voice typing), gallery mein "car" likho to car photos, X-ray se diagnosis.



$AI \supset ML \supset DL$

— DL, ML ka tareeqa; ML, AI ka hissa

🧠 **Yaad rakho:** Russian dolls 🪆: bahar AI, andar ML, sab se andar DL. DL ko ML se bahar nahi nikal sakte.

5.1.1 Machine Learning (ML)

Def (paper mein yehi likho): *A machine learning model is the system that learns from data and gives an output.*

1. Data input → 2. Learning (patterns) → 3. Prediction (naya data)

Normal programming	Machine learning
Har rule khud likho: `if marks >= 40: "Pass"`	Rules list nahi karte — data (examples) do, model khud pattern seekhe
Input + rules → output	Input + output (examples) → rules/model

🧠 **Yaad rakho:** **D-L-P**: "Data Learns to Predict." Model = program ka trained dimagh.

5.1.3 Deep Learning (DL)

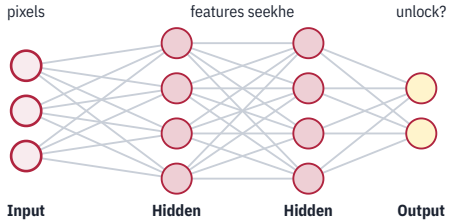
Def (paper mein yehi likho): *Deep learning is a machine-learning method in which computers use multi-layer neural networks to learn complex patterns from data.*

Shuru ki layers **saada features** (edges, shapes, rang); gehri layers unhein jor ke **complex** cheez (poora chehra, Cultus vs Mehran).

	Neural network (model)	Deep learning (method)
Kya	Layers, nodes, connections wala system	Bohat hidden layers wale NN ko train karne ka tareeqa
Gehrai	Shallow ho sakta (1–2 hidden)	Hamesha deep; bohat data + power
Role	Architecture ("machine")	Training ka tareeqa

5.1.2 Neural Networks (NN) ★ Golden

Def (paper mein yehi likho): *A neural network is a computer model that learns patterns from data.*



Layer	Kaam (face unlock)
Input	Raw data leta — chehre ke pixels
Hidden	"Sochta" — features seekhta (aankhon ka faasla, naak)
Output	Final jawab — unlock / locked

🧠 **Yaad rakho:** **I-H-O**: "In, Hidden, Out" — input leta, hidden sochta, output faisla. Dimagh se *inspired* maths machine — asli dimagh ki copy nahi.

📝 **Paper mein:** "What is a neural network?" (2–3) · "3 layers face-unlock misaal ke saath" (5) · NN vs DL.

5.1.4 Components of Neural Networks ★ Golden

#	Component	Kaam
1	Input layer	Raw data (pixels, sound, text)
2	Neurons / nodes	Chhoti units: lein, hisaab, aage bhejein
3	Hidden layer(s)	Beech mein process, patterns seekhein
4	**Weights**	Connection ki **ahmiyat/taqat**
5	**Bias**	Extra constant — nateeja adjust (dhakka)
6	**Activation function**	Math gate: info aage jaye ya nahi
7	Output layer	Final result ("This is a cat")
8	**Loss function**	Ghalti kitni → network khud theek kare

$$s = x_1 w_1 + x_2 w_2 + b, \qquad y = f(s) = \begin{cases} 1 & s \geq 0.5 \\ 0 & s < 0.5 \end{cases}$$

Worked:

$$x_1 = 0.8, x_2 = 0.5, w_1 = 0.4, w_2 = 0.6, b = 0.1$$

→

$$s = 0.32 + 0.30 + 0.1 = 0.72 \geq 0.5$$

→

$$y = 1$$

.

🧠 **Yaad rakho:** Weights = ahmiyat · Bias = extra nudge · Activation = haan/na gate · Loss = kitna ghalat.

📝 **Paper mein:** "List 8 components" + "weights & bias kya?" + numerical neuron (x, w, b, threshold diya ho).

Drill: (1) AI, ML, DL mein sab se bara set? (2) ML ke 3 steps? (3) NN ki chhoti processing units? (4) "Deep" kyun? (5) Weights kya dikhate?

Jawab: 1. AI · 2. Data input, learning, prediction · 3. Neurons/nodes · 4. Kai hidden layers · 5. Neurons ke connection ki ahmiyat/taqat | **5-mark:** Neural Networks.

Day 17

DL Components, Pakistan, Applications & Secure Collaboration

Components of DL · Future in Pakistan · Applications of NN & DL · Authentication, MFA, Access Control

★ **Must lock:** Secure Collaboration (Authentication, MFA, Access Control)

5.1.5 Components of Deep Learning

#	Component	Matlab
1	**D**ata (large input)	"Data hungry" — hazaron/lakhon records
2	Many hidden **L**ayers	Darjanon/saikron layers — gehre patterns
3	**A**utomatic feature learning	Insaan features na bataye — khud nikale
4	**L**earning algorithm	Backpropagation — ghaltiyan kam
5	High computing power	**G**PUs (Graphics Processing Units)
6	**T**raining process	Bar bar loops = **epochs**

🧠 **Yaad rakho: D-L-A-L-G-T**: "Deep Learners Always Love Good Training."

Future of AI, NN, DL in Pakistan

National AI Policy 2025 — AI ab qaumi planning ka hissa: innovation, skills, ethics, digital growth.

Sector	Mumkin asar
Education	Smart learning, online tutoring, auto assessment
Healthcare	Bimari detect, medical images — dehaat mein specialist kam
Agriculture	Fasal monitoring, mausam, irrigation, paidawar
Business & banking	Fraud detection, chatbots, forecasting
Local languages	Urdu, Sindhi ki speech recognition, translation, TTS

🧠 **Yaad rakho: E-H-A-B-L**.

5.2 Applications of NN & DL

Application	Misaal
Image recognition	Face unlock; Karachi plaza camera staff list se milaye
Speech recognition	Alexa se alarm; Urdu voice search
NLP	Google Translate English → Urdu; bank chatbot
Healthcare	Lung X-ray mein pneumonia/tumour pattern (DHQ hospital)
Recommendation	YouTube programming tutorials suggest kare
Autonomous vehicles	Self-driving car pedestrian dekh ke brake
Fraud detection	HBL/Meezan app: naye shehar se raat 3 baje Rs 80,000 flag
Assistive tech	Speech-to-text; Urdu screen reader

🧠 **Yaad rakho: I-S-N-H-R-A-F-A**.

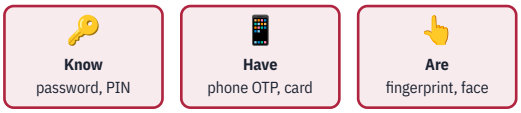
5.3.1 Secure Collaboration ★ Golden

Def (paper mein yehi likho): *Secure collaboration means using digital tools to work together in a safe, controlled and responsible way.*

Files share, online classes, documents mil ke edit — faida tabhi jab data mehfooz ho.

Element	Sawal / matlab	Misaal
Authentication	"**Tum kaun ho?**" — pehchan verify	Password, fingerprint, MFA
Access control	"**Tum kya kar sakte ho?**" — permissions	Teacher edit, student sirf view
Accountability	Kis ne kya kiya — record	Activity logs, version history
Privacy	Sirf zaroori logon se share	"Anyone with link" band
Safe transmission	Raste mein data encrypted	HTTPS, encrypted chat

MFA (Multi-Factor Authentication) = kam az kam **do alag qisam** ke factors. **2FA** = exactly do (password + OTP).



⚠ **Dhoka:** Do passwords = 2FA nahi (dono "know" qisam ke). Factors **alag qisam** ke hone chahiye.

🧠 **Yaad rakho: A-A-A-P-S**. Authentication = darwaze pe pehchan; access control = andar kaunse kamre khulein.

📝 **Paper mein:** "Define secure collaboration", "authentication vs access control", "What is MFA/2FA?", "5 elements with examples".

5-mark frame: Neural Network

Define	A computer model that learns patterns from data.
Explain	Input → hidden → output layers; weights, bias, activation, loss.
Example	Face unlock: pixels → features → unlock/locked.
Diagram	3 columns of circles joined by lines.
Working	$s = \Sigma\, x \cdot w + b$ → activation → y.

Drill: (1) DL "data hungry" kyun? (2) Pakistan ki AI policy? (3) Bola hua lafz → text? (4) Authentication aur access control kaunsa sawal?

Jawab: 1. Bohat bare datasets chahiye · 2. National AI Policy 2025 · 3. Speech recognition · 4. Kaun ho? / Kya kar sakte ho? | **5-mark:** Secure Collaboration.

Day 18

Data Protection, Security Threats & Mitigation

Encryption · Passwords · 3-2-1 Backup · Firewall · Malware, Phishing, DoS · Identifying · Mitigation

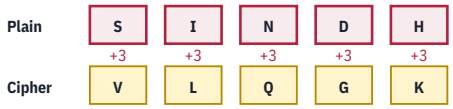
★ **Must lock:** Common Security Threats (Malware, Phishing, DoS)

5.3.2 Data Protection

Def (paper mein yehi likho): *Data protection means keeping digital information safe from unauthorised access, misuse, loss, damage or accidental deletion.*

Tareeqa	Kaam
Encryption	Data ko na-parhne-qabil code mein (sirf key wala parhe)
Strong passwords	Lamba, mix (A-z, 0-9, @#), share nahi
Backup (3-2-1)	3 copies, 2 media types, 1 off-site (cloud)
Firewall	Network ka gate — mashkook traffic block
Antivirus	Andar talaashi — malware dhoond ke hatao

Caesar cipher (shift 3): har harf 3 aage.



PAK → **SDN** (P→S, A→D, K→N). Decrypt = 3 peeche.

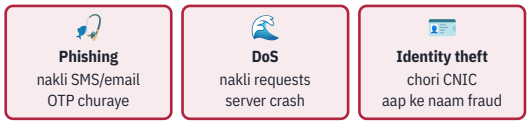
🧠 **Yaad rakho:** Cipher = paigham pe taala · Password = darwaze pe taala · 3-2-1 = teen jagah chaabiyan · Firewall = gate · Antivirus = andar talaashi.

5.4.2 Identifying Threats

Nishani	Shayad
Slow PC, files ghaib, random pop-ups	Malware
Failed logins, ajeeb login location	Unauthorised access
Jaldi karo! OTP do, bad grammar, ajeeb link	Phishing
Records achanak delete/copy/move	Data theft / insider
Server requests ka achanak toofan	DoS attack

5.4.1 Common Security Threats ★ Golden

Def (paper mein yehi likho): *A security threat is any action, event or weakness that can harm a computer system or its data. Mitigation techniques are protective measures that prevent threats or reduce their impact.*



Threat	Kya hai	Misaal
Malware	Nuqsan-deh software: viruses, worms, spyware, ransomware, Trojans	Cracked software; "free game" USB lab ka coursework lock kare
Phishing	Social engineering — nakli email/site se password, OTP	JazzCash/HBL jaisa SMS "verify karo"
Unauthorised access	Bina ijazat system/account	Khula lab PC, doosre ka Classroom
Data / identity theft	Data chori; identity = kisi ki details se uska roop	Leaked CNICs se SIM/wallet
DoS	Hazaron nakli requests → slow/crash	BISE result day pe portal flood
Human error	Jaiz users ki ghalti	Marksheet ghalat parent ko email

🧠 **Yaad rakho:** Malware = bura program · Phishing = nakli paigham · DoS = **toofan** · Identity theft = chori hua "aap" · Human error = imaandar ghalti.

📝 **Paper mein:** "Define security threat", "malware/phishing/DoS with examples", "data theft vs identity theft" — chhe ke chhe threats likho.

5.4.3 Mitigation Techniques

Technique	Kaise madad
Antivirus + Firewall	USB malware hatao; internet traffic filter
Strong passwords + 2FA	Guessing mushkil; OTP extra layer
Encryption	Intercept ho bhi to parha na jaye
Regular backups	Crash, ghalti, ransomware ke baad wapsi
User awareness	Anjaan links/attachments se bacho — insaan sab se kamzor *aur* sab se mazboot kari

🧠 **Yaad rakho:** Paanch dhaalein: **AF · P2 · E · B · U**. Ransomware files lock kare → backup hi bachata.

Drill: (1) PAK ko Caesar shift 3? (2) Malware ki 5 qismein? (3) Random pop-ups + files ghaib? (4) 2FA password ke upar kaise?

Jawab: 1. SDN · 2. Viruses, worms, spyware, ransomware, Trojans · 3. Malware · 4. Phone pe OTP — doosra factor | **5-mark:** Security Threats.

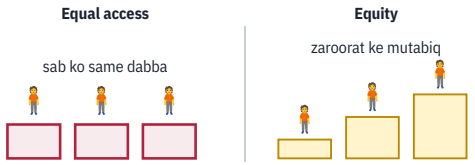
Equity, Equal Access, Collaboration Tools & Careers

Equity vs Equal Access · Functions · Collaboration Tools · AI & Cyber Careers

★ **Must lock:** Functions of Equity and Equal Access

5.5 Equity & Equal Access

Collaboration tools online learning ke liye zaroori — har koi **bina rukawat** hissa le.



Equal access	Equity
Sab ko **bilkul same** moqa, tools, resources	Har ek ko **zaroorat ke mutabiq extra madad** taake nateeja barabar
Sab ko same joote	Wheelchair wale ke liye ramp

5.5.1 Functions of Equity & Equal Access ★ Golden

Function	Kya karta	Pakistani misaal
1. Need-based **S**upport	Har fard ke liye khaas madad	Nabina student ke liye screen reader / audio descriptions (Karachi)
2. **R**emoving barriers	Door-daraz students online class mein	Low-data, mobile-friendly apps — Tharparkar, katcha areas
3. Active **P**articipation	Live class chhoot jaye to bhi hissa	Recorded lectures — Hyderabad mein load-shedding
4. Different learning needs (**F**ormats)	Ek se zyada formats	Same topic ka video, notes, audio

🧠 **Yaad rakho:** "**S**ome **R**ural **P**upils **F**ail" — agar ye na hon: Support, Remove, Participate, Formats.

📝 **Paper mein:** Golden 5–8 marks: chaar headings tarteeb se, har ek ki misaal + equal vs equity ka farq.

5-mark frame: Equity & Equal Access

Define	Equal access gives everyone the same opportunity, tools and resources; equity gives extra need-based support for a fair outcome.
Explain	4 functions: support, remove barriers, participation, formats.
Example	Screen reader; low-data app in Tharparkar; recorded lecture.
Diagram	Same boxes vs different-height boxes.
Working	Teacher LMS pe video + notes + audio deta hai.

5.5.2 Collaboration Tools

Tool	Kaam / taleemi use
Google Docs	Real-time mil ke likhna, comment — group assignments
Zoom	Online meetings, virtual class, screen share
Microsoft Teams	Meetings + files + class groups ek jagah
Google Classroom / LMS	Lectures, assignments, quizzes, grading
WhatsApp	Foran announcements — zimmedari se
Email	Formal notices, bari attachments, professional record

🧠 **Yaad rakho:** Docs = saath likho · Zoom/Teams = milo · Classroom = official hub · WhatsApp = tez magar informal · Email = formal record.

Careers (AI & Cybersecurity)

Career	Kaam
Machine Learning Engineer	ML/NN models banaye, train kare (face recognition, NLP)
Cybersecurity Analyst	Malware/phishing pe nazar; firewall, 2FA
AI Researcher	Naye AI/DL techniques — university labs, R&D
Data Privacy Officer	Data handling privacy qawaneen ke mutabiq
Ethical Hacker	Qanooni tor pe kamzoriyan dhoonde (penetration testing)

Chapter 5 Exam Radar 🎯

- AI ⊃ ML ⊃ DL; NN 3 layers; 8 components; neuron numerical
- Secure collaboration 5 elements; authentication vs access control; MFA
- 6 threats + examples; identify signs; 5 mitigations; Caesar cipher
- Equity vs equal access; 4 functions; tools

Checkpoint: (1) Equal access ek jumle mein? (2) Equity ke 4 functions? (3) Real-time group writing tool? (4) Network ki kamzoriyan qanooni tor pe kaun jaanchta?

Jawab: 1. Sab ko bilkul same moqa, tools, resources · 2. Need-based support, removing barriers, active participation, different learning needs · 3. Google Docs · 4. Ethical hacker | 60% se kam → kal Golden repair.

Day 20

Entrepreneurship, Digital Age & Prototype

Entrepreneur · Digital Age · Local Examples · Problem to Idea · Prototype

★ **Must lock:** Entrepreneurship in the Digital Age, Prototype

6.1 Entrepreneur & Entrepreneurship

Def (paper mein yehi likho): *An entrepreneur is a person who identifies a problem, thinks of a solution, and tries to change that solution into a product or service.*

1. Problem → 2. Solution → 3. Product / service

- Logon ko dekhte, unki zaroorat sunte — kya bar bar tang karta hai.
- **Calculated risk** lete — kya kho sakte, nuqsan chhota kaise rakhein, soch ke.
- Idea qadam ba qadam behtar — pehli koshish perfect nahi.

Entrepreneur	Entrepreneurship
Shakhs	**Process** — plan, start, test, improve

🧠 **Yaad rakho:** Entrepreneur poochta: "Main kaunsa masla hal kar sakta hoon?"

Local Examples (Table 6.1)

Problem	Digital idea	Pehle users
Parents college notices miss karte	WhatsApp notice + reminder service	First-year students ke parents
Embroidery shop ke customers kam	Facebook/WhatsApp catalogue + order form	Isi shehar ke customers
Students test dates bhoolte	Google Calendar guide / reminder app	Grade XII students
Canteen pe break mein lambi line	Snacks ka pre-order form	Students & teachers

🧠 **Yaad rakho:** "**Problem pehle, tool baad mein.**" Takleef se shuru karo, technology se nahi.

6.3 Problem → Business Idea

Chaar sawal: Masla kya? Kis ko? Kitni baar? Abhi log kya karte?

Step	Students kya karein	Misaal
Observe	Ghar/college/bazaar mein bar bar ka masla	Practical file dates bhoolna
Ask users	Mutasir logon se baat	5 classmates se poochho
Select one problem	Sirf ek saaf masla	Practical file reminder
Suggest solution	Saada digital/kaghazi hal	Reminder form / app screen
Test early	Final se pehle users ko dikhao	"Use karoge?"

🧠 **Yaad rakho:** "**O**nly **A**sk **S**imple **S**olutions **T**oday".

6.2 Entrepreneurship in the Digital Age ★ Golden

Def (paper mein yehi likho): *Entrepreneurship in the digital age means using digital tools to start, test, improve, and grow a business idea.*

Tools: mobile phones, social media, websites, online forms, digital payments, maps, videos, simple apps. Pakistan mein: WhatsApp, Facebook, Instagram, TikTok, YouTube, Google Forms, Google Maps, JazzCash/Easypaisa.

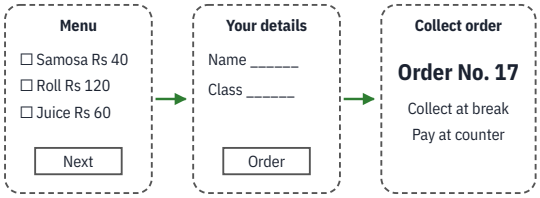
Pehle	Ab (digital)
Dukaan, kiraya, bara kharcha	WhatsApp/Facebook page — takreeban muft
Sirf mohalle ke customers	Poore shehar/mulk tak
Feedback lena mushkil	Google Form, comments — foran
Test karna mehenga	Sasta, tez test

🧠 **Yaad rakho:** **S-T-I-G**: Start, Test, Improve, Grow — "Start The Idea Growing."

📝 **Paper mein:** "Define entrepreneurship in the digital age", "Digital tools ne kaise badla?", "Bina bari investment ki misaalein".

6.4 Prototype ★ Golden

Def (paper mein yehi likho): *A prototype is an early model of a product, service, website, or app that shows how the idea may look or work; it is NOT the final product and is made to test the idea and collect feedback.*



Canteen pre-order ka paper prototype: item chuno → naam/class → order collect

Faide
Idea jaldi aur sasta dikhao · ghaltiyan pehle pakro · user feedback lo · waqt/paisa bachao · team ka idea saaf

🧠 **Yaad rakho:** Prototype = "practice model" — essay ka rough draft: dikhao, correction lo, final se pehle dobara likho.

📝 **Paper mein:** 3–5 marks: definition (*early model, not final, test idea, collect feedback*) + faide + misaal (paper screens).

Drill: (1) Entrepreneurship shakhs ya process? (2) 4 digital tools? (3) Embroidery shop ka idea? (4) Table 6.2 ke 5 steps? (5) Prototype final product?

Jawab: 1. Process · 2. Phones, social media, online forms, digital payments · 3. FB/WhatsApp catalogue + order form · 4. Observe, Ask, Select, Suggest, Test early · 5. Nahi — test aur feedback ka early model | **5-mark:** Prototype.

Day 21

Prototype Types, Development Cycle & MVP

Types of Prototypes · Development Cycle · ClassBuddy Activity · MVP · Prototype vs MVP

★ **Must lock:** Prototype Development Cycle, MVP

6.5 Types of Prototypes (Fidelity)

Fidelity = prototype final product se kitna qareeb.



Type	Matlab	Best use	Misaal
Low-Fi	Jaldi, kacha model	Bohat early test	Haath se bani homework-reminder screens
Mid-Fi	Layout + flow saaf	Navigation test	PowerPoint/Figma clickable screens
High-Fi	Final ke qareeb, tafseeli	Asli tajurba test	Order form website ka working demo

🧠 **Yaad rakho:** Fidelity = "wafadari". Paper → PowerPoint → Polished.

6.7 Class Activity: "ClassBuddy"

Students homework deadlines, test dates, class items bhoolte. Team poora app foran nahi banati:

1. Samjho → 2. Low-fi draw → 3. Test → 4. Feedback analyse → 5. Behtar

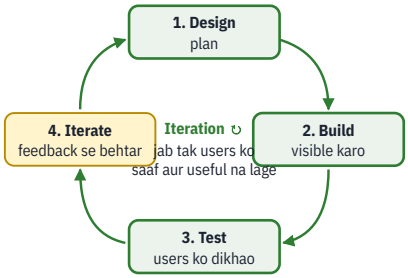
Finding	Matlab	Design decision
10 mein se 7 bhoolte (70%)	Masla aam hai	Daily task list dikhao
Ghar se nikalne se pehle reminder	Waqt ahem	Subah notification option
Saada app chahiye	Design saada	Bare buttons, saaf labels

$$\frac{7}{10} \times 100 = 70\%$$

🧠 **Yaad rakho:** Feedback → Problem → Improvement: har comment ko masle mein, phir ek saaf tabdeeli mein.

6.6 Prototype Development Cycle ★ Golden

Ek dafa ka kaam nahi — banao, test, behtar, phir test. Is dohrane ko **iteration** kehte hain.



Stage	Kya hota
1. Design	Masla samjho, hal plan — prototype kya dikhaye (screens, features)
2. Build	Pehla version (physical/digital) — maqsad khoobsurti nahi, idea **nazar** aaye
3. Test	Users ko dikhao — kya samjhe, kahan mushkil
4. Iterate	Feedback se behtar — faltu hatao, kami jodo → wapas Design

🧠 **Yaad rakho:** "**D**o **B**ig **T**hings **I**teratively". Aakhri arrow wapas Design — **cycle** hai, line nahi.

📝 **Paper mein:** Long Q: "Explain the prototype development cycle with a diagram" — 4 boxes circle mein, arrows, har stage 1–2 lines.

6.8 Minimum Viable Product (MVP) ★ Golden

Def (paper mein yehi likho): *A Minimum Viable Product (MVP) is the simplest working version of a product.*

Minimum = sirf zaroori features, extras nahi. **Viable** = itna kaam kare ke asal masla hal ho ("kamyabi se chal sake").

Poori canteen app (final)	Canteen MVP
Login, online payment, delivery tracking, discounts...	Saada Google Form — break se pehle snacks chuno

🧠 **Yaad rakho:** MVP jawab deta: "Kya log **waqai** use karenge?" Design nahi, asal behaviour test.

6.9 Prototype vs MVP

Point	Prototype	MVP
Maqsad	Idea kaisa dikhe/chale — **design** test	Log waqai use karenge? — **market demand** test
Working	Chalna zaroori nahi — sketch/model	Kam az kam basic tor pe **chale**
Users	Chhota group design pe feedback	Asal early users rozmarra mein
Cost	Bohat kam	Kam, magar thora development
Misaal	Canteen ke paper screens	Google Form 3 din asal orders ke liye

🧠 **Yaad rakho:** Prototype = "Sahih dikhta/samajh aata?" · MVP = "Log use karenge aur faida payenge?" **Market demand** = kaafi log chahte aur use/paise denge?

Drill: (1) Fidelity? (2) Iteration? (3) Step 1 mein kitne bhoole, %? (4) MVP full form? (5) Kaunsa basic tor pe chalna zaroori?

Jawab: 1. Final product se kitna qareeb · 2. Banao, test, behtar, dobara test ka dohrao · 3. 7/10 = 70% · 4. Minimum Viable Product · 5. MVP | **5-mark:** Prototype Cycle / MVP.

Riskiest Assumption, Developing & Testing an MVP

Riskiest Assumption · Developing an MVP · Canteen Case Study · Testing an MVP

★ **Must lock:** None — tables copy karo

6.10 Riskiest Assumption

Assumption = jo hum sach maante magar test nahi kiya. **Riskiest assumption** = ghalat nikle to **poora business** doob jaye. Pehle test karo — waqt aur paisa bachta.

Business idea	Risky assumption	MVP se test
Online tuition finder	Students tutoring ke paise denge	20 students, trial class — sign up?
Canteen pre-order	Break se pehle order karenge	Google Form ek hafta asal waqt mein
Ajrak online catalogue	WhatsApp pe order denge	30 customers ko catalogue, asal orders gino
College van updates	Roz updates dekhenge	Ek route ka trial WhatsApp broadcast

🧠 **Yaad rakho:** Poochho: "Kya cheez is idea ko **maar** degi?" — wohi riskiest. Sasta, asal behaviour se pehle test.

6.11 Developing an MVP

Sirf zaroori features; shuru mein extras nahi. Aam ghalti: sab kuch ek saath banana → mushkil, mehenga, slow.

MVP element	Sawal	Misaal
Problem	Kaunsa masla?	Canteen pe lamba intezar
Target users	Pehle kaun?	Ek college ke Grade XII
Main feature	EK feature jo hal kare?	Break se pehle pre-order
Basic design	Kaunse steps?	Menu, naam, class, waqt
Cost	Muft/sasta tool?	Google Form / printed slips
Marketing	Users ko pata kaise?	Class WhatsApp group
Feedback	Jawab kaise?	3 din baad chhota form
Improvement	Kya badlein?	Feedback ke mutabiq

🧠 **Yaad rakho:** "**P**eople **T**ry **M**y **B**akery; **C**heap **M**uffins **F**ind **I**mprovement".

6.12 Case Study: Canteen Pre-Order MVP

Lambi lines; kabhi counter tak pohanchte snacks khatam. Maqsad: poora app **nahi** — idea test.

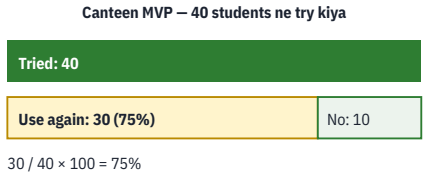
Paper prototype → Feedback: chhota menu, collection pe payment → Google Form MVP → 3-din test → Behtar

Element	Faisla
Features	Menu, naam, class, item, quantity
Cost & tools	Muft Google Form ya printed slips
Marketing	Class announcement + WhatsApp
Testing period	Teen din
Feedback	Waqt bacha? Dobara use karoge?
Improvement	Sirf 5 items; break se 20 min pehle orders band

🧠 **Yaad rakho:** "**5 items, 3 days, 20 minutes**".

6.13 Testing an MVP

Basic version asal users ko do, response parho. **Numbers** (quantitative) = kitno ne try kiya/dobara karenge; **comments** (qualitative) = kya pasand/na-pasand/uljhan. Plan: kitne users, kitne din.



Feedback type	Sawal	Nateeja
Number-based	Kitno ne try kiya?	40 students
Number-based	Dobara use?	40 mein se 30 haan
Comment-based	Kya uljhan thi?	Order closing time maloom nahi
Observation	Kahan mushkil?	Class ka naam bhool gaye

🧠 **Yaad rakho:** Feedback tabhi faida jab parha jaye: **gino → dohrate masle dhoondo → pehli improvement chuno**.

5-mark frame: Minimum Viable Product (MVP)

Define	The simplest working version of a product, with only essential features, usable enough to solve the main problem.
Explain	Minimum = sirf zaroori; Viable = kaam kare. Asal users ke saath market demand test.
Example	Canteen: Google Form, 3 din, 40 students, 30 dobara.
Diagram	Final app (bohat features) vs MVP (ek feature).
Working	Riskiest assumption chuno → 8 elements → test → feedback → improve.

Feedback se faisla — misaal

Feedback	Masla	Tabdeeli
"Closing time maloom nahi tha"	Info ghaib	Form ke upar: "Orders break se 20 min pehle band"
Class naam bhool gaye	Field optional	Class field **required**
Menu lamba	Faisla mushkil	Sirf 5 items
Paisa online nahi	Payment rukawat	Counter pe payment

Drill: (1) Assumption? (2) Sab ek saath banana ghalti kyun? (3) Canteen MVP ka tool aur muddat? (4) Table 6.10 ke 3 feedback types?

Jawab: 1. Jo sach maante magar test nahi kiya · 2. Mushkil, mehenga, slow · 3. Google Form, 3 din · 4. Number-based, comment-based, observation

Beachhead Market, Project, Ethics & Careers

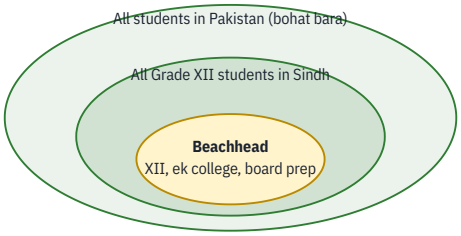
Beachhead Market · Student Project Rubric · Ethical Use · Startup Careers

★ **Must lock:** Beachhead Market

6.14 Beachhead Market ★ Golden

Def (paper mein yehi likho): *A beachhead market is a small and focused group of first users who have a clear need and are likely to try the product early.*

Sab ko serve karne ki bajaye **ek chhota group** pehle → MVP theek test, behtar, phir phailao.



Andar wale chhote circle se shuru, seekho, behtar karo, phir bahar phailo

Idea	Kamzor pehla market	Behtar beachhead
Study app	Pakistan ke sab students	Ek college ke XII, exam prep
Canteen pre-order	Shehar ki sab canteens	Ek college canteen, lunch break
Ajrak catalogue	Sab online shoppers	Ek local shop ke purane customers
Van updates	Sab transport users	EK van route ke students/parents

🧠 **Yaad rakho: S-R-U**: **S**pecific (tang), **R**eachable (team pohanch sake), **U**rgent need. Hafte mein pehle users se na mil sako → market bohat bara.

📝 **Paper mein:** "Define beachhead market + characteristics" ya "Suggest a beachhead for ..." — definition + S-R-U + kamzor vs behtar misaal.

6.16 Ethical & Safe Use

Value	Matlab
Privacy	Logon ki zaati info unki — ijazat ke baghair share nahi
Truthfulness	Nateeje imaandari se — numbers na banao, bura feedback na chhupao
Fairness	Sab users se barabar sulook
Equal access	Har koi use kar sake

🧠 **Yaad rakho: P-T-F-E**: Private, Truthful, Fair, Equal-access?

6.15 Complete Student Project

Groups mein chhota, local project. Safar: **problem → prototype → feedback → MVP plan → risky assumption → beachhead**.

Ideas: homework/test reminder, college notice organizer, canteen pre-order, dukaan ka WhatsApp catalogue.

Criteria	Excellent	Satisfactory	Needs work
Problem ID	Saaf, local, ahem	Samajh aaye, detail kam	Mubham / bohat bara
Prototype	Saaf flow + features	Basic idea	Adhoora
Feedback & fixes	Sahih collect; tabdeeli feedback se	Analysis kamzor	Feedback nahi
MVP plan	Zaroori features + test plan	Basic, testing ghair-wazeh	Final product se confuse

🧠 **Yaad rakho:** Submit se pehle rubric row-by-row: "Kya ye Excellent hai?" Feedback & Fixes mein likho **kaunse** feedback se **kaunsi** tabdeeli.

Careers (Startups)

Pakistan mein incubators: **National Incubation Centres (NICs)**.

Career	Kaam
Tech Entrepreneur / Founder	Masla → digital product: prototype → MVP → launch
Product Manager	User needs, features ki tarjeeh, prototype/MVP test
Business Analyst	Masail analyse, digital hal, assumptions test
Innovation Consultant	Ideas validate, MVP, beachhead mein mashwara
Digital Marketing Specialist	MVP ko beachhead tak pohanchana, users barhana
Venture / Incubator Analyst	Early startups ko investment/incubation ke liye jaanchna

Chapter 6 Exam Radar 🎯

- Entrepreneur vs entrepreneurship; digital age definition (S-T-I-G)
- Prototype: define, types (fidelity), cycle diagram
- MVP: Minimum + Viable; prototype vs MVP table
- Riskiest assumption; 8 MVP elements; canteen case
- Beachhead: define + S-R-U + misaal

5-mark frame: Beachhead Market

Define	A small, focused group of first users with a clear need who are likely to try the product early.
Explain	Sab ki bajaye ek group → MVP theek test → phir phailao. S-R-U: specific, reachable, urgent.
Example	Canteen app: ek college ki canteen, lunch break.
Diagram	Concentric circles — andar beachhead.

Checkpoint: (1) Achhe beachhead ki 3 khoobiyan? (2) 4 project ideas? (3) Feedback lete waqt 3 values? (4) MVP ko beachhead tak kaun pohanchata?

Jawab: 1. Specific, reachable, urgent need · 2. Reminder system, notice organizer, canteen pre-order, shop catalogue · 3. Privacy, truthfulness, fairness · 4. Digital Marketing Specialist | 60% se kam → kal Golden repair.

Day 24

Golden Blitz — Chapters 1 & 2

Sirf ★ cards · 11 Golden · har table aur diagram yaad se likho, phir check

★ Must lock: 11 Golden cards (Ch 1–2)

Pehle har card scrap paper pe **yaad se** likho (table + diagram), phir milao. Sahih ho to ☐ pe ✔. Ghalat walay Day 29 repair list mein.

☐ Applications of HCI

Ch1 · 1.2

HCI designs interactive systems (apps, websites) as human-centred solutions across domains of life.

H-BEN	Application
Health	telemedicine, portals
Banking	mobile app, ATM
Education	LMS, online exams
Networking	WhatsApp, Zoom

☐ Importance of HCI

Ch1 · 1.4

HCI makes machines approachable and usable — complex tech into everyday tools.

P-I-I-E-E: Productivity, Inclusivity, Innovation (AR/VR, AI, IoT), Economic impact, Ethical responsibility. Bura design = medical device ke ghair-wazeh buttons.

☐ HCI Problems

Ch1 · 1.7

Complexity (checkout na mile) · Poor feedback (Submit freeze) · Inconsistency (🔍 vs 🔭) · Accessibility gaps (no captions) · Compatibility (mobile crash) · Contextual neglect (no dark mode) · Cognitive load (3 SMS codes) · Trust deficiency (no HTTPS)

☐ UI vs UX

Ch1 · 1.9

UI design creates and organises visual elements so users interact easily.

	UI	UX
Kya	Look: buttons, icons, colors	Poora tajurba / feeling
Misaal	Outfit	Poora din
Tool	Figma	Research, testing

Components: buttons, menus, icons, text fields, checkbox/radio, sliders.

☐ Testing Methods

Ch1 · 1.11.2

Observing how the system behaves under specific conditions.

Method	Key
Usability	real users, real tasks
Task-based	time · success · errors
A/B	2 versions, 50/50, zyada clicks jeete
Automated audit	bot: links, speed, contrast

Evaluation: SUS (10 sawal), NPS (recommend?), formative vs summative.

☐ Trace Tables

Ch2 · 2.1.1

Monitors how variable values change during execution.

Dry run · debugging · validation. Columns: step, variables, condition, output.

height	1.57–1.91?	Total
1.91	T	1
2.01	F	1
1.87	T	2
1.57	T	3
1.48	F	3

Output: **3**

☐ Trace vs Stepwise

Ch2 · 2.1.3

	Trace	Stepwise
Best for	chhote blocks, beginners	bare systems, proofs
Effort	high	moderate
Level	micro, exact values	macro, overall state

☐ Big O

Ch2 · 2.3.1

Describes how running time grows as input size n grows.

Class	Misaal	Grade
$O(1)$	arr[0]	Excellent
$O(\log n)$	binary search	Great
$O(n)$	linear search	Fair
$O(n \log n)$	merge sort	Good
$O(n^2)$	nested loops	Poor

$$3n + 5 \Rightarrow O(n)$$

· n = 16: 1, 4, 16, 64, 256

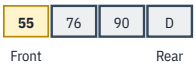
☐ Data Structure

Ch2 · 2.5

Systematic way of organising, storing and managing data for efficient access and processing.

Linear: array, linked list, stack, queue · Non-linear: tree, graph.





☐ Stack (LIFO)

Ch2

PUSH/POP at **TOP**. PUSH D1, D2, D3 → POP = **D3**. Undo, back button.

☐ Queue (FIFO)

Ch2

ENQUEUE at **Rear**, DEQUEUE from **Front**. 55, 76, 90, D → DEQUEUE = **55**. Printer jobs.

Blitz score

Ch1 ☐☐☐☐☐ / 5
Ch2 ☐☐☐☐☐☐ / 6

Day 25

Golden Blitz — Chapters 3 & 4

Sirf ★ cards · 11 Golden · code aur tables yaad se likho

★ Must lock: 11 Golden cards (Ch 3–4)

Pehle har card scrap paper pe **yaad se** likho (table + diagram), phir milao. Sahih ho to ☐ pe ✔. Ghalat walay Day 29 repair list mein.

☐ Lists — Ch3 · 3.1.1

Ordered, mutable collection in [ ].

append, insert, remove (pehla), pop, sort, reverse, index, count. `L[-1]` aakhri; `nums[2:6]` → [2,3,4,5].

```
f = ["kiwi", "apple"]
f.append("grape")
f.remove("apple")
f.sort()
```

☐ Tuples — Ch3 · 3.1.2

Ordered, immutable collection in ( ).

	List	Tuple
Mutable?	Haan	Nahi
Methods	bohat	count, index
Speed	slow	tez, safe

`t[0] = 99` → TypeError

☐ Dictionaries — Ch3 · 3.1.4

Key–value pairs; ordered (3.7+), mutable, unique keys.

`d["age"] = 31` · `del d["city"]` · `d.get(k, 0)` (crash nahi) · keys(), values(), items(). Repeat key → aakhri value.

```
freq[w] = freq.get(w, 0) + 1
```

☐ Functions — Ch3 · 3.2

Named block written once, run whenever called.

def → name → (params) → body → return. Benefits: reusability, modularity, readability, maintainability. DRY.

```
def add(a, b):
    return a + b
print(add(2, 3))   # 5
```

☐ File Handling r/w/a — Ch3 · 3.3

Create, read, write, update, delete files on disk (persistent).

Mode	Kaam	File na ho
r	parho	error
w	mita ke likho ⚠	nayi
a	aakhir mein jodo	nayi

read(), readline(), readlines(), write(), close()

☐ with (Context Manager) — Ch3 · 3.6

Opens the file, runs the block, always closes it — even on error.

```
with open("f.txt", "a") as f:
    f.write("David,81\n")
```

try / except FileNotFoundError / else / finally

☐ DB Connection — Ch4 · 4.1.2

Database → Driver (sqlite3) → Connection (pipe) → Cursor (messenger) → SQL.

connect → cursor → execute → commit → fetchall → close

```
c = sqlite3.connect("a.db")
cur = c.cursor()
cur.execute(sql)
c.commit()
rows = cur.fetchall()
c.close()
```

☐ Pandas DataFrame — Ch4 · 4.3

2-D table of rows and columns.

```
import pandas as pd
df = pd.DataFrame(data)
df["Math"].mean()
```

Name	Math	Science
Ali	85	80
Sara	90	88
Ahmed	78	70

Math mean = 84.33

☐ Missing Values (NaN) — Ch4 · 4.3.3

`isnull().sum()` gino · `fillna(mean)` numbers · `fillna("Unknown")` categories · `dropna()` rows hatao. Pehle saaf, phir analyse.

☐ Graph Types — Ch4 · 4.5

Chart	Kab
Bar/column	categories
Line	time trend
Pie	whole ke hisse
Histogram	distribution, bars jurri
Scatter	2 variables
Box	spread, outliers

Title + axis labels zaroori.

☐ Descriptive Statistics — Ch4 · 4.6

$$\bar{x} = \frac{\sum x}{n}$$

· median = sorted beech · mode = sab se zyada · range = max − min ·

$$SD = \sqrt{\text{variance}}$$

2,4,6,8,10: mean 6, median 6, range 8, var 8 (n) / 10 (n−1), SD 2.83 / 3.16

Blitz score

Ch3 ☐☐☐☐☐☐ / 6
Ch4 ☐☐☐☐☐ / 5

Golden Blitz — Chapters 5 & 6

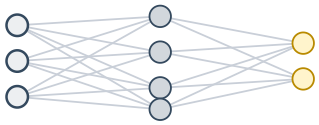
Sirf ★ cards · 10 Golden · definitions word-for-word bolo

★ **Must lock:** 10 Golden cards (Ch 5–6)

Pehle har card scrap paper pe **yaad se** likho (table + diagram), phir milao. Sahih ho to ☐ pe ✔. Ghalat walay Day 29 repair list mein.

☐ Neural Networks — Ch5 · 5.1.2

A computer model that learns patterns from data.



Input (pixels) → Hidden (features) → Output (unlock?). I-H-O.

$$AI \supset ML \supset DL$$

.

☐ NN Components — Ch5 · 5.1.4

Input · neurons · hidden · **weights** (ahmiyat) · **bias** (nudge) · **activation** (gate) · output · **loss** (ghalti).

$$s = 0.8(0.4) + 0.5(0.6) + 0.1 = 0.72 \geq 0.5 \Rightarrow y = 1$$

☐ Secure Collaboration — Ch5 · 5.3.1

Using digital tools to work together safely, controlled and responsibly.

A-A-A-P-S: Authentication (kaun?), Access control (kya kar sakte?), Accountability, Privacy, Safe transmission. MFA = know + have + are.

☐ Security Threats — Ch5 · 5.4.1

Threat	Misaal
Malware	virus, worm, ransomware
Phishing	nakli SMS, OTP chori
Unauthorised access	khula PC
Data/identity theft	chori CNIC
DoS	requests ka toofan
Human error	ghalat email

Shields: antivirus+firewall, password+2FA, encryption, backup, awareness.

Nishaniyan: pop-ups → malware · ajeeb login → access · "OTP do!" → phishing · traffic spike → DoS. Caesar +3: PAK → SDN.

☐ Equity & Equal Access — Ch5 · 5.5.1

Equal = sab ko same; Equity = zaroorat ke mutabiq madad.

Function	Misaal
Support	screen reader
Remove barriers	low-data app
Participation	recorded lecture
Formats	video + notes + audio

☐ Digital Entrepreneurship — Ch6 · 6.2

Using digital tools to start, test, improve and grow a business idea.

S-T-I-G · WhatsApp, Forms, JazzCash. Problem pehle, tool baad mein.

☐ Prototype — Ch6 · 6.4

Early model to test the idea and collect feedback — not the final product.

Low-Fi (paper) → Mid-Fi (PowerPoint/Figma) → High-Fi (demo).

☐ Prototype Cycle — Ch6 · 6.6



Design → Build → Test → Iterate ↻ wapas Design. "Do Big Things Iteratively".

☐ MVP — Ch6 · 6.8

Simplest working version of a product.

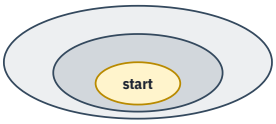
Minimum = zaroori features; Viable = kaam kare. Canteen: Google Form, 3 din, 5 items, 20 min.

	Prototype	MVP
Maqsad	design test	market demand
Chale?	zaroori nahi	basic tor pe haan
Misaal	paper screens	Form 3 din

Riskiest assumption pehle test.

☐ Beachhead Market — Ch6 · 6.14

Small, focused group of first users with a clear need.



S-R-U: Specific, Reachable, Urgent. Ek college ke XII students, phir phailo.

Blitz score

Ch5 ☐☐☐☐☐ / 5

Ch6 ☐☐☐☐☐ / 5

Day 27

Night-before Recaps — Saare Chapters

Kitaab band karo · har recap ounchi awaaz mein bolo · jo na bol sako, mark karo

★ **Must lock:** All chapters

Ch 1 Human-Computer Interaction

- HCI = log aur computer saath kaam karein — usable, useful, satisfying. HCI = CHI = MMI.
- Traditional = keyboard, mouse; natural = touch, voice, gesture, face.
- Components: user, computer, interaction, task (+ interface, environment, feedback).
- UI = look/controls; UX = poora tajurba. Wireframe (Lo-Fi → Hi-Fi) → prototype (Figma) → test.
- Evaluate: usability, task-based, A/B, automated audit, SUS, NPS. Accessibility C-F-C-N-M.

Ch 2 Algorithms & Data Structures

- Correctness: trace table (har step values) ya stepwise reasoning (logic blocks).
- Clarity: modularity (SRP) + readability (N-S-S-C).
- Efficiency: steps, conditions, repetitions. Big O:

$$1 < \log n < n < n \log n < n^2$$

.

- Linear: array, linked list, stack (LIFO), queue (FIFO). Non-linear: tree (root), graph (nodes + edges).
- Undo → stack · printer → queue · hierarchy → tree · maps → graph · fast index → array.

Ch 3 Python Data Structures, Functions, Files

- list [ ] mutable · tuple ( ) immutable · set { } unique · dict {k:v}.
- def, parameters, return; local vs global; built-in vs user-defined.
- Files: r read, w overwrite, a append; `with open(...) as f` khud close.
- try / except / else / finally — specific error pehle.

Ch 4 Data and Analysis

- SQLite: connect → cursor → execute → commit → close. Writes ke baad commit!
- DataFrame = table; read_csv, isnull, dropna, fillna, sort, filter, groupby.
- Bar (categories), line (time), pie (whole), histogram (distribution), scatter (2 vars), box (outliers).
- Mean, median, mode, range, variance, SD; pandas .std() n−1.

Ch 5 AI, Security, Collaboration

- AI ⊃ ML ⊃ DL; neuron = weighted sum + bias → activation. Deep = bohat hidden layers.
- Authentication (MFA) + access control; encryption, passwords, 3-2-1 backup, firewall.
- Threats: malware, phishing, DoS, identity theft, human error.
- Equity: devices, zubaan, disability, bandwidth — sab hissa lein.

Ch 6 Entrepreneurship

- Entrepreneur = masla → venture, calculated risk. Digital tools = sasta test.
- Prototype: low/mid/high fidelity; Design → Build → Test → Iterate.
- MVP = riskiest assumption asal users se test. Prototype ≠ MVP.
- Beachhead = pehla tang group. Data ethically (P-T-F-E).

Mnemonic wall 🧠

HCI = CHI = MMI | S-T-H-V-S | H-BEN | U-C-I-T | P-I-I-E-E | C-F-C-N-M | Time · Success · Errors | N-S-S-C | Constant Logs Never Need Squares | S-B-W | I-D-T-S-S | LIFO / FIFO | DRY | r · w · a | P-D-A-G | D-L-P | I-H-O | D-L-A-L-G-T | A-A-A-P-S | Some Rural Pupils Fail | S-T-I-G | Do Big Things Iteratively | S-R-U | P-T-F-E

Paper ke pakke sawal 🎯

- HCI applications (H-BEN) · importance (P-I-I-E-E) · problems (8) · UI vs UX · testing methods
- Trace table dry-run · trace vs stepwise · Big O table · stack/queue trace · scenario → structure
- List methods/slicing · list vs tuple · dict get · functions benefits · r/w/a + with + try-except
- DB connection 5 parts · SQLite steps · NaN handling · which graph · mean/median/SD
- NN layers + components + neuron sum · secure collaboration · threats · equity functions
- Digital entrepreneurship · prototype + cycle · MVP vs prototype · beachhead (S-R-U)

Aage ka plan: Day 28 — mock, hall conditions. **Day 29** — key se mark, har missed ★ ki repair list. **Day 30** — sirf repair list; raat 10 baje so jao; koi naya topic nahi.

Day 28

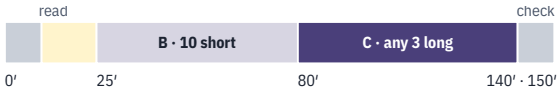
Mock Paper — Hall Conditions (1/2)

2 hours 30 minutes · 75 marks · no notes · Section A

★ **Must lock:** Whole syllabus

Aaj ke 3 ghante — Hall conditions

- **2 h 30 min** mein paper · **75 marks** · notes nahi, phone doosre kamre mein, Day 29 ke pages **mat dekho** (answers wahan hain — un pages ko alag rakh do).
- Bacha hua 30 min: paper pack karo — **aaj mark nahi karna**.
- Khud apne invigilator bano: timer lagao, ghari saamne, ek hi baith mein.



Time plan (minutes): pehle 10′ paper parho aur Section C ke 3 sawal chuno · A 15′ · B 55′ (~5′ har sawal) · C 60′ (20′ har sawal) · aakhri 10′ check

Paper strategy

- MCQ: pehle yaqeeni wale, shak wale pe nishaan, baad mein.
- Short (3 marks): **Define + 2 points/example**.
- Long (10 marks): **Define · Explain · Example · Diagram · Working**.
- Har diagram/table pe **label + title**.
- Python: sahih **indentation**, colon, quotes.

Mock paper — Computer Science XII

Time: 2 h 30 min · Total: 75 · Calculators allowed

Instructions: Answer all of Section A and B; attempt any **three** from Section C. Write Python with correct indentation. Draw neat, labelled diagrams.

Section A — Multiple choice

15 × 1 = 15

1. HCI stands for:
   a. Human Computer Interaction b. High Code Interface
   c. Hardware Control Input d. Hybrid Computing Internet
2. A wireframe is:
   a. the final coloured UI b. a simple layout sketch of screens
   c. a compiled program d. an encryption key
3. Which is $O(\log n)$?
   a. Linear search b. Nested for-loops
   c. Binary search d. Reading one array cell
4. A stack is:
   a. FIFO b. LIFO
   c. random access only d. unordered unique items
5. Which Python type is immutable?
   a. list b. dict
   c. set d. tuple
6. The file mode that overwrites an existing file is:
   a. r b. a
   c. w d. x+
7. `with open("t.txt") as f` is useful because:
   a. it encrypts the file b. it closes the file automatically
   c. it creates a DataFrame d. it skips exceptions
8. After INSERT in sqlite3 you must:
   a. rollback always b. commit
   c. drop the table d. print the cursor
9. A pie chart is a poor choice when:
   a. you have 3 parts of a total b. you have 20 tiny slices
   c. you show percentages of a whole d. you label each slice
10. pandas `df['x'].std()` uses:
    a. population n b. sample n−1
    c. median d. mode
11. Deep Learning is a subset of:
    a. only statistics b. Machine Learning
    c. only HCI d. only networking
12. MFA means:
    a. Many File Archives b. Multi-Factor Authentication
    c. Main Firewall Access d. Minimum Function API
13. A Caesar cipher is a form of:
    a. phishing b. encryption
    c. a firewall d. a prototype
14. An MVP is:
    a. the most polished version of a product b. the smallest product that tests the riskiest assumption
    c. a paper sketch only d. a full company registration
15. A printer job list is best modelled as a:
    a. stack b. queue
    c. binary tree d. set

MCQ answer grid — sahih harf pe gola lagao

1. ⓐ ⓑ ⓒ ⓓ	**2.** ⓐ ⓑ ⓒ ⓓ	**3.** ⓐ ⓑ ⓒ ⓓ	**4.** ⓐ ⓑ ⓒ ⓓ	**5.** ⓐ ⓑ ⓒ ⓓ	**6.** ⓐ ⓑ ⓒ ⓓ	**7.** ⓐ ⓑ ⓒ ⓓ
8. ⓐ ⓑ ⓒ ⓓ	**9.** ⓐ ⓑ ⓒ ⓓ	**10.** ⓐ ⓑ ⓒ ⓓ	**11.** ⓐ ⓑ ⓒ ⓓ	**12.** ⓐ ⓑ ⓒ ⓓ	**13.** ⓐ ⓑ ⓒ ⓓ	**14.** ⓐ ⓑ ⓒ ⓓ
15. ⓐ ⓑ ⓒ ⓓ						

Kal (Day 29) isi grid ko key se milao — overwriting / do gole = ghalat.

Day 28

Mock Paper — Hall Conditions (2/2)

Section B (short) · Section C (long, any three)

★ **Must lock:** Whole syllabus

Section B — Short questions

10 × 3 = 30

1. Differentiate UI and UX with one example. [3]
2. What is a trace table? Why is it better than "it looks right"? [3]
3. Draw a 3-node stack after push 10, push 20, pop. Label TOS. [3]
4. Write one difference each: list vs tuple vs set vs dictionary. [3]
5. Write a Python function `area(l, w)` that returns l*w, and a call that prints it. [3]
6. Name the five steps of a database connection in Python. [3]
7. When would you draw a scatter plot rather than a line graph? [3]
8. Compute mean, median and range of 4, 8, 6, 10, 2. [3]
9. List three signs of a phishing message. [3]
10. Differentiate a prototype and an MVP. [3]

Section C — Long questions

Attempt any THREE · 3 × 10 = 30

1. A college wants a new student-portal. Describe need analysis, a low-fidelity wireframe of the login + timetable screens, three accessibility principles you would apply, and one usability test you would run. [10]
2. Trace this algorithm with a table for n = 5: total = 0; for i in 1..n: total = total + i; return total. State its Big O and rewrite it as a formula that is $O(1)$. [10]
3. Write a Python program that reads a text file of lines "name,marks", stores results in a dictionary, and writes "name: grade" to an output file using `with`. Grade A if marks ≥ 80 else B. Include exception handling if the file is missing. [10]
4. Using the dataset 12, 18, 12, 20, 15, compute mean, median, mode, range, population variance and population standard deviation, showing every step. Then state which chart you would use to show the distribution and why. [10]
5. A Karachi canteen wants a pre-order app. Identify the problem, the riskiest assumption, an MVP, a beachhead market, two test metrics, and one ethical rule for student data. [10]

Rough-work / time log

Section	Shuru	Khatam	Kitne sawal
A			/15
B			/10
C			/3

Marks ↔ jawab ki lambai

Marks	Kitna likho
1 (MCQ)	Sirf sahih harf
3 (short)	Definition + 2 points / 4–6 lines ya ek chhota diagram
10 (long)	1½–2 pages: Define · Explain · Example · Diagram · Working

Section C — mera intikhab

Choice	Q no.	Kyun (strongest topic)
1st		
2nd		
3rd		

Paper khatam? 🛑

Kalam rakho. Paper ek lifafe mein band. **Aaj marking nahi** — kal fresh dimagh se Day 29 key ke saath. Raat ko bas aaram.

🧠 **Yaad rakho:** Long answer ka frame: **Define · Explain · Example · Diagram · Working** — kisi ek ke baghair poore marks nahi.

Day 29

Mark the Mock — Section A & B Key

0–90 min: key se mark karo · ghalat sawal ka Golden note karo

★ **Must lock:** Missed Goldens

⚠ **Dhoka:** Pehle **Day 28 ka paper poora** ho chuka ho — warna ye page mat parho. Jis jawab mein working, diagram ya example nahi, usay **poore marks mat do**.

Section A — key (15)

Q	Ans	Ghalat ho to repair	✔/✘
1	**a**	1.1 HCI fundamentals	☐
2	**b**	1.9.3 Wireframing	☐
3	**c**	★ Big O	☐
4	**b**	★ Stack	☐
5	**d**	★ Tuples	☐
6	**c**	★ File handling (r, w, a)	☐
7	**b**	★ with statement	☐
8	**b**	★ DB connection (commit)	☐
9	**b**	★ Data visualization	☐
10	**b**	★ Descriptive statistics	☐
11	**b**	5.1 AI ⊃ ML ⊃ DL	☐
12	**b**	★ Secure collaboration	☐
13	**b**	5.3.2 Data protection	☐
14	**b**	★ MVP	☐
15	**b**	★ Queue	☐

A score: ____ / 15

Section B — model answers (10 × 3)

Q	Model answer (in points ko dhoondo)	Repair topic	Mine
1	UI = screens & controls (buttons, rang). UX = poora safar (waqt, frustration, trust). JazzCash send screen (UI) vs payment kitni mehfooz/jaldi lagi (UX).	★ UI vs UX	/3
2	Har instruction ke baad har variable ki value. Chune hue input pe correctness saabit; "looks right" off-by-one aur uninitialised variables miss karta.	★ Trace tables	/3
3	 push 10 → TOS 10; push 20 → TOS 20; pop → 20 gaya, TOS = 10.	★ Stack	/3
4	list: ordered, mutable [ ] · tuple: ordered, immutable ( ) · set: unordered, unique { } · dict: key:value pairs.	★ Lists/Tuples/Dicts	/3
5	`def area(l, w): return l * w` · `print(area(3, 4))` → 12	★ Functions	/3
6	connect → cursor → execute → commit → close.	★ DB connection	/3
7	Scatter: do measured variables (height vs marks), time axis nahi. Line: ek quantity waqt/sequence ke saath.	★ Data visualization	/3
8	Sorted 2, 4, 6, 8, 10 · mean $30/5 = 6$ · median 6 · range $10 - 2 = 8$.	★ Descriptive statistics	/3
9	Anjaan sender · urgent dhamki ("account aaj band") · link bank se match na kare · OTP/password maange · bura Urdu/English · anjaan attachment (koi 3).	★ Security threats	/3
10	Prototype = seekhne ka model (aksar internal). MVP = asli, usable product — bas itne features ke riskiest assumption asli customers se test ho.	★ Prototype / MVP	/3

B score: ____ / 30 · Marking: 1 mark definition/sahih term, 1 mark explanation, 1 mark example/diagram.

Section A — har jawab ki wajah

Q	Kyun yahi jawab? (MCQ wazahat)
1	HCI = Human-Computer Interaction (CHI / MMI bhi).
2	Wireframe = Lo-Fi layout sketch, rang nahi.
3	Binary search har step aadha karta hai.
4	Stack = Last-In First-Out.
5	Tuple ( ) read-only — badal nahi sakte.
6	'w' pehle mita deta, phir likhta.
7	Block khatam → file khud band.
8	INSERT/UPDATE/DELETE disk pe tabhi jab commit.

Q	Kyun yahi jawab? (MCQ wazahat)
9	20 chhote slices — pie parhna na-mumkin; bar behtar.
10	pandas .std() default sample (n−1).
11	AI ⊃ ML ⊃ DL — DL, ML ka hissa.
12	Multi-Factor Authentication: know / have / are.
13	Caesar = letters shift = encryption.
14	MVP = riskiest assumption test karne wala sab se chhota product.
15	Printer: pehle aaya pehle print → FIFO queue.

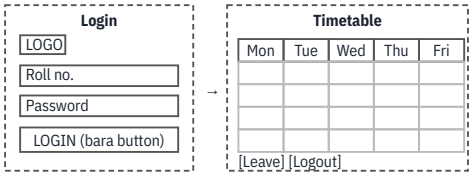
Mark the Mock — Section C Worked Solutions

Har long answer ka model · marks ka hisaab

★ **Must lock:** Missed Goldens

C1 — Student portal (HCI) [10] ★ Golden

Need analysis	Jawab
Users	Students, teachers, office
Tasks	Timetable dekhna, leave submit
Constraints	Mobile, slow internet, Urdu/English



Accessibility (koi 3): bare tap targets · colour akela signal nahi (icon/text bhi) · Urdu/English labels · screen-reader names · achha contrast.

Usability test: 5 students **think-aloud** karte hue login + timetable kholein; **time, errors, success** note.

Marks: need analysis 3 · wireframe 3 · accessibility 3 · test 1

C2 — Trace + Big O [10] ★ Golden

i	1	2	3	4	5
total	1	3	6	10	**15**

Loop body n dafa →

$$O(n)$$

. Formula:

$$\text{total} = \frac{n(n+1)}{2} = \frac{5\times6}{2} = 15$$

→

$$O(1)$$

(n kitna bhi bara, ek hi hisaab).

```python
def total_loop(n):          # O(n)
    total = 0
    for i in range(1, n + 1):
        total = total + i
    return total

def total_formula(n):       # O(1)
    return n * (n + 1) // 2
```

Marks: trace table 4 · output 1 · Big O + wajah 2 · formula + O(1) 3

Section C marking guide

- Definition/sahih terms ke baghair → max 6/10.
- Wireframe/table bina label → −2.
- Python indentation/colon ghalat → −1 to −3.
- Stats: har step (mean → deviations → squares → ÷n → √) na dikhaya → −3.
- Sirf 2 long sawal? Teesre ke 0.

C3 — Dictionary + files + exception [10] ★ Golden

```python
grades = {}
try:
    with open("marks.txt") as f:
        for line in f:
            name, m = line.strip().split(",")
            grades[name] = int(m)
except FileNotFoundError:
    print("Error: marks.txt is missing")
else:
    with open("grades.txt", "w") as g:
        for name, m in grades.items():
            grade = "A" if m >= 80 else "B"
            g.write(name + ": " + grade + "\n")
```

Sample: `Ali,85`, `Sara,72` → dict `{'Ali': 85, 'Sara': 72}` → grades.txt: `Ali: A`, `Sara: B`.

C4 — Statistics 12, 18, 12, 20, 15 [10] ★ Golden

x	$x - \bar{x}$	$(x - \bar{x})^2$
12	−3.4	11.56
12	−3.4	11.56
15	−0.4	0.16
18	2.6	6.76
20	4.6	21.16
Σ 77	0	**51.2**

Sorted 12, 12, 15, 18, 20 · mean

$$77/5 = 15.4$$

· median 15 · mode 12 · range 8 ·

$$\sigma^2 = 51.2/5 = 10.24$$

.

$$\sigma = \sqrt{10.24} = 3.2$$

. Chart: **histogram** (distribution) ya frequencies ka bar.

C5 — Karachi canteen pre-order [10] ★ Golden

Element	Jawab
Problem	Periods ke darmiyan lambi canteen queues
Riskiest assumption	Students break se 10 min pehle pre-order karenge
MVP	WhatsApp number + Google Form, pickup tokens — app nahi
Beachhead	First-year students ki ek shift
Metrics	% orders pick-up · time-to-collect · repeat use
Ethics	Phone numbers publish nahi; term ke baad responses delete

Aam ghaltiyan — Section C

Q	Ghalti → sahih
C1	Sirf "achha design" → users/tasks/constraints + labelled wireframe
C2	Big O bina wajah → "loop n dafa" likho; formula $O(1)$
C3	open bina with; int() bhoolna; except na lagana
C4	Variance ko "marks" mein; n vs n−1 na batana
C5	MVP = poori app likhna → sab se chhota test

C score: ____ / 30

Day 29

Score, Repair List & Golden Checklist

90–150 min: repair list bharo · 150–180 min: ek missed Golden yaad se

★ **Must lock:** Missed Goldens

Score tracker

Section	Max	Mine	%
A — MCQ	15		
B — Short	30		
C — Long	30		
Total	**75**		

$$\text{Percentage} = \frac{\text{marks}}{75} \times 100 \qquad \text{e.g. } \frac{54}{75} \times 100 = 72\%$$

Score	Matlab	Kal (Day 30)
≥ 80% (60+)	Tayyar ✔	Sirf repair list, jaldi so jao
60–79% (45–59)	Theek, kuch kamzor	Repair list poori, har item 2 dafa
< 60% (< 45)	Serious repair	Sirf top 6 Goldens — naya kuch nahi

Chapter-wise kamzori

Ch	A ghalat	B kam	C kam	Priority
Ch 1				
Ch 2				
Ch 3				
Ch 4				
Ch 5				
Ch 6				

Repair list ✍

Har Golden jis pe **0 ya aadhe** marks aaye. Kal **sirf yahi** — poori kitaab nahi.

#	Golden topic missed	Main yaad se kya likhunga (def / table / diagram)	Done
1			☐
2			☐
3			☐
4			☐
5			☐
6			☐
7			☐
8			☐
9			☐
10			☐

150–180 min: in mein se **ek** Golden abhi yaad se dobara likho.

Golden checklist — mock mein jo miss hua us pe ✔ (32 Goldens)

Chapter	Goldens (miss hua to ✔)	Missed
Ch 1	☐ Applications of HCI ☐ Importance of HCI ☐ HCI Problems ☐ UI vs UX ☐ Testing Methods	__ / 5
Ch 2	☐ Trace Tables ☐ Trace vs Stepwise ☐ Big O ☐ Data Structure concept ☐ Stack ☐ Queue	__ / 6
Ch 3	☐ Lists ☐ Tuples ☐ Dictionaries ☐ Functions ☐ File handling (r, w, a) ☐ with statement	__ / 6
Ch 4	☐ DB connection ☐ Pandas & DataFrames ☐ Missing values (NaN) ☐ Data visualization ☐ Descriptive statistics	__ / 5
Ch 5	☐ Neural Networks ☐ NN components ☐ Secure collaboration ☐ Security threats ☐ Equity functions	__ / 5
Ch 6	☐ Digital entrepreneurship ☐ Prototype ☐ Prototype cycle ☐ MVP ☐ Beachhead market	__ / 5

Day **30**

Repair Day — Phir Ruk Jao

Sirf Day 29 ki repair list · 6 pm stop · raat 10 baje neend

★ **Must lock:** Missed Goldens only

Aaj ka plan (3 ghante · 6 baje band)

Waqt	Kaam
0–10′	Day 29 ki repair list kholo — sirf woh items
10–100′	Har missed Golden pe **3-step redo** (neeche)
100–115′	Break — paani, thora chalo; phone nahi
115–170′	Doosra round: **kitaab band**, har item ounchi awaaz mein + ek table/diagram
170–180′	Bag pack — bas! **6 pm stop**

⚠ **Dhoka:** Naya chapter nahi · naya non-Golden card nahi · aakhri raat "sab kuch dobara" nahi.

3-step redo (har missed Golden)

1. Definition bolo (word-for-word) → 2. Table / diagram yaad se banao → 3. Ek 5-mark answer likho

#	Golden	Def	Table/Diag	5-mark
1		☐	☐	☐
2		☐	☐	☐
3		☐	☐	☐
4		☐	☐	☐
5		☐	☐	☐
6		☐	☐	☐
7		☐	☐	☐
8		☐	☐	☐

Raat ki checklist 🎒

☐ Admit card / roll no. slip ☐ 2 blue/black pens + spare
☐ Pencil, sharpener, eraser ☐ Ruler (diagrams, tables)
☐ Calculator (agar allowed) ☐ Paani ki bottle
☐ Ghari (hall clock pe bharosa nahi) ☐ Kapre tayyar ☐ Alarm set
☐ Raat 10 baje so jao 😴

Goldens **dimagh mein** le kar jao — booklet hall mein nahi.

Hall mein — 6 sunehri usool

1. Pehle 10′ poora paper parho; Section C ke apne **3 strongest** sawal chuno.
2. MCQ pehle — jaldi, magar har option parh ke.
3. Har long answer: **Define · Explain · Example · Diagram · Working**.
4. Diagrams pencil + ruler se, **labels aur title** ke saath.
5. Python: indentation, colons, quotes check.
6. Aakhri 10′ — roll no., question numbers, khali chhore sawal check.

5-mark frame — ek nazar 🧠

Step	Kya likho	Misaal (Stack)
Define	Kitaab wali exact English definition	A stack is a linear data structure that works on the LIFO principle: Last-In, First-Out.
Explain	2–3 lines Urdu soch, English likhai	PUSH top pe add, POP top se nikale; TOP sab se upar.
Example	Pakistani / rozmarra misaal	Undo button; canteen ki plates.
Diagram	Labelled diagram ya table	Vertical boxes, TOP label, PUSH/POP arrows.
Working	Steps / trace / formula	PUSH D1, D2, D3 → POP = D3 → stack [D1, D2].

Aakhri nazar — 6 pm se pehle ek dafa 👀

Topic	Aakhri nazar (mnemonic / formula)
HCI	H-BEN domains · P-I-I-E-E importance
Accessibility	C-F-C-N-M
UI vs UX	UI = jo dikhe; UX = jo mehsoos ho
Big O	$O(1) < O(\log n) < O(n) < O(n \log n) < O(n^2)$
Stack / Queue	LIFO push/pop · FIFO rear in, front out
Python types	[ ] list · ( ) tuple · { } set · {k:v} dict

Topic	Aakhri nazar (mnemonic / formula)
Files	r read · w wipe+write · a add · with = auto close
SQLite	connect → cursor → execute → commit → close
Stats	$\bar{x} = \frac{\sum x}{n}$, $\sigma = \sqrt{\frac{\sum (x-\bar{x})^2}{n}}$
Neural network	I-H-O; weights, bias, activation, loss
Security	A-A-A-P-S · MFA know/have/are · 3-2-1 backup
Startup	Do Big Things Iteratively · MVP · beachhead S-R-U

🧠 **Yaad rakho:** 30 din mukammal. Tum ne har Golden kam az kam 3 dafa dekha hai. Ab **aaram** — kal behtareen do. 🌟 Best of luck!